

MAHIDOL UNIVERSITY

Faculty of Information and Communication Technology



Lecture 3
Data and Signals

August 2026

ปัญญาของแผ่นดิน Wisdom of the Land

มหาวิทยาลัยมหิดล
Mahidol University

ห้ามมิให้นักศึกษาทำซ้ำ ดัดแปลง หรือใช้ประโยชน์จากงานอันมีลิขสิทธิ์ปรากฏอยู่ในระบบการศึกษาอิเล็กทรอนิกส์ (E-Learning) ของมหาวิทยาลัย ไม่ว่าจะทั้งหมดหรือบางส่วน โดยไม่ได้รับอนุญาตจากมหาวิทยาลัย นอกเหนือจากการศึกษาส่วนบุคคล ทั้งนี้ การทำซ้ำ ดัดแปลง หรือเผยแพร่ต่อสาธารณชนซึ่งงานอันมีลิขสิทธิ์ **จะมีโทษปรับตั้งแต่ 20,000 บาท ถึง 200,000 บาท และหากเป็นการกระทำเพื่อการค้า จะมีโทษจำคุกตั้งแต่ 6 เดือน ถึง 10 ปี หรือปรับตั้งแต่ 100,000 บาท ถึง 800,000 บาท หรือทั้งจำทั้งปรับ**

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Quote of the Week



Someone is sitting in the shade today because someone planted a tree a long time ago.

Warren Buffet

PAY IT **FORWARD** ☺

Terminologies before a break

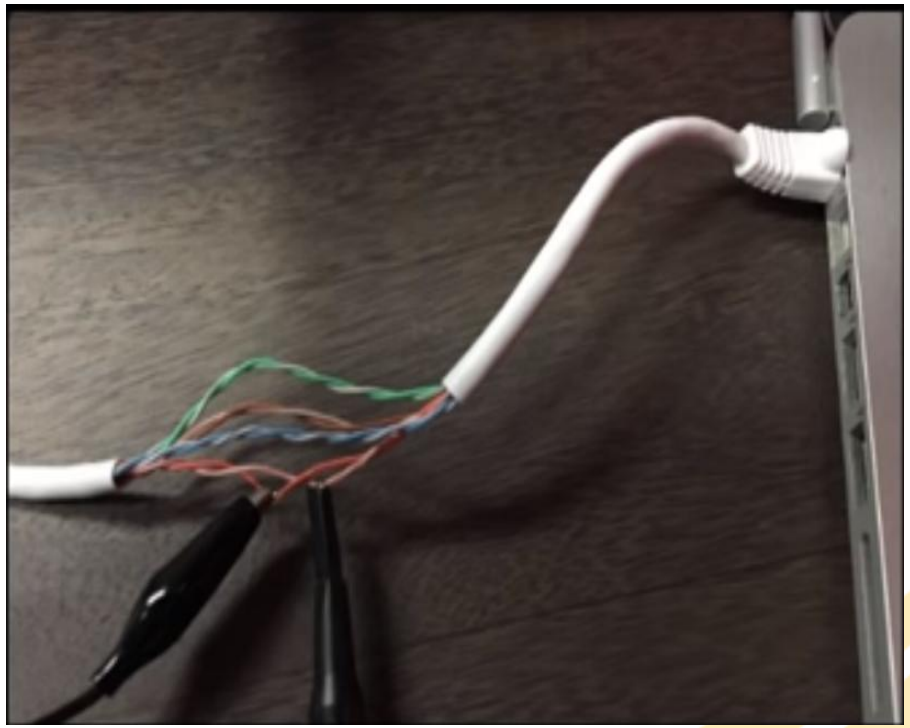
1. Oscilloscope
2. Aliasing Effect
3. Sampling
4. Minimum sampling rate
5. Nyquist theorem
6. Time-domain graph
7. Frequency-domain graph
8. Square-wave signal
9. Bandwidth
10. Data Rate & Signal Rate



Signaling Measurement

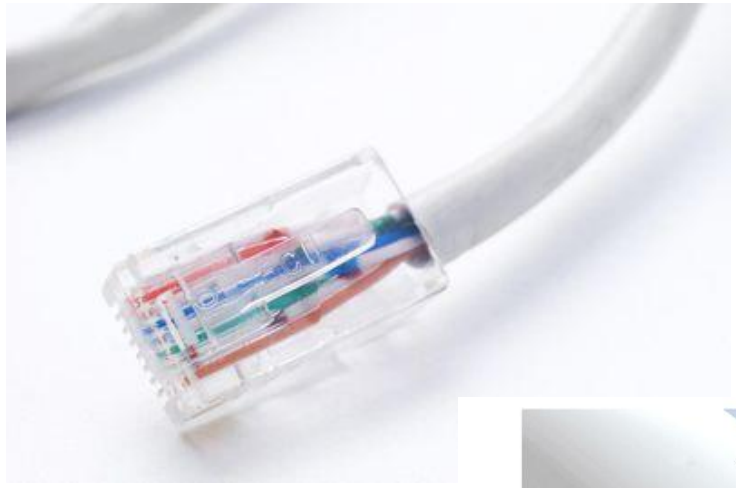
- We can measure signals (usually electrical signals) using a device called Oscilloscope.
- In this case, the transmission of data signal over Ethernet cable
- Setup:

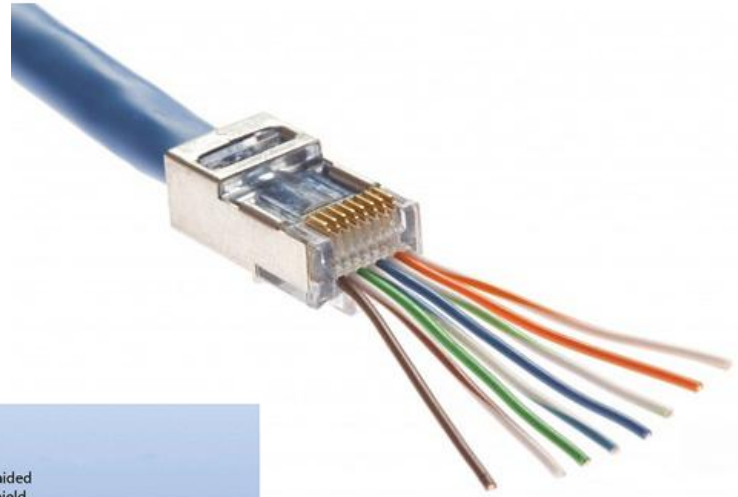


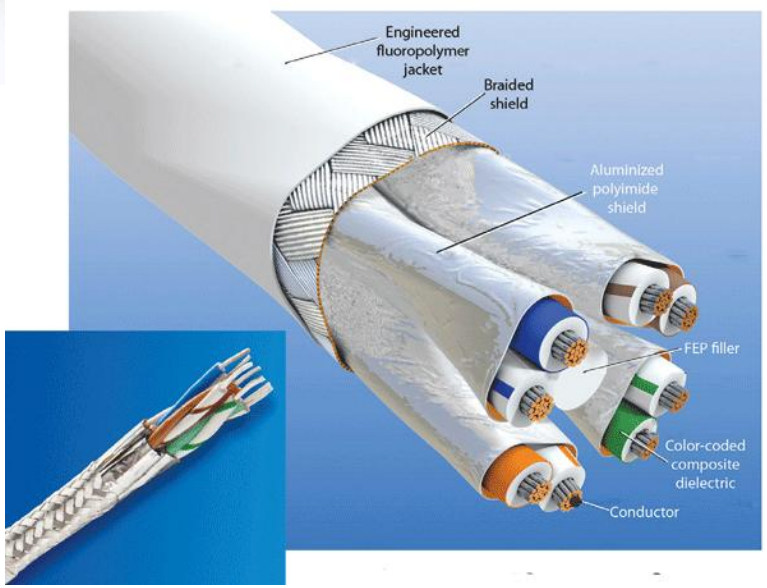


Ethernet Cable

Ethernet cable or LAN cable has many types. Popular types are CAT-5e and CAT-6







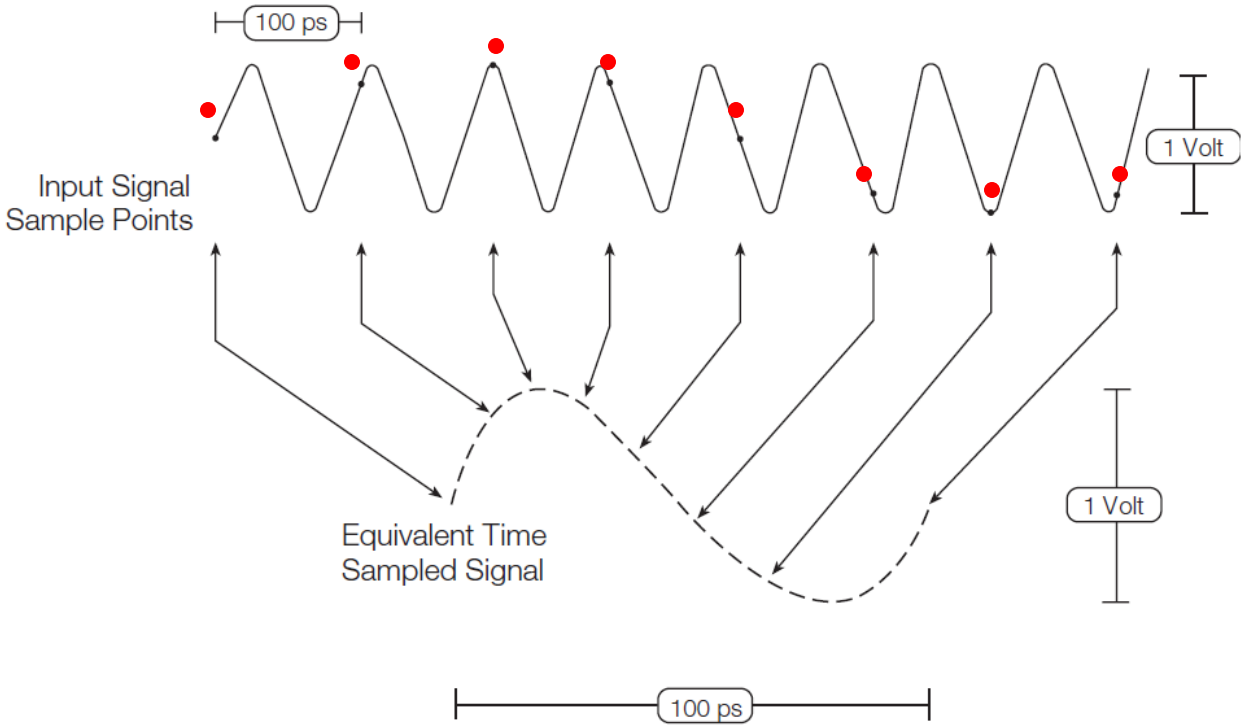
Oscilloscope

- previously called an oscillograph, and informally known as a scope or o-scope, is a type of electronic test instrument that graphically displays varying signal voltages, usually as a two-dimensional plot of one or more signals as a function of time.
- An oscilloscope displays a graph showing the voltage snapshots of an electrical signal over time, with **voltage as the Y-axis** and **time as the X-axis**.



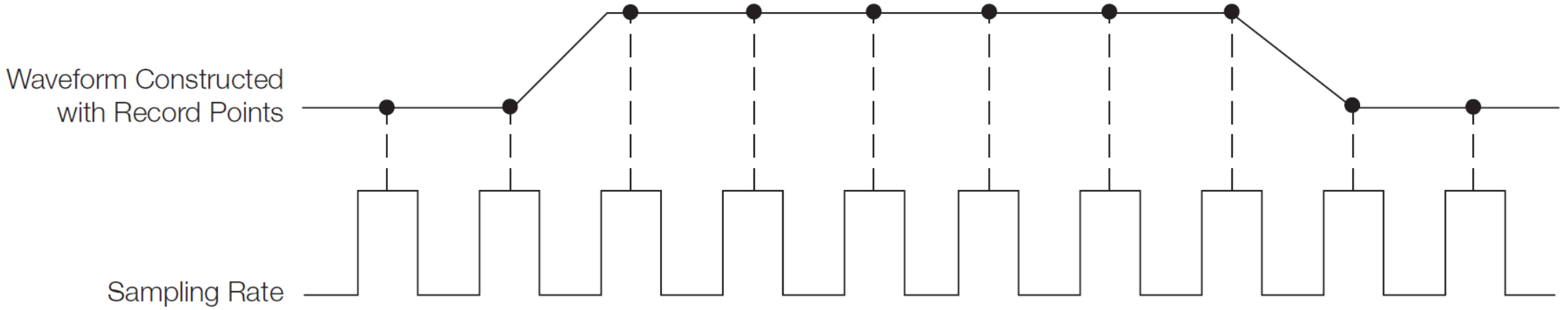
Sampling

- It is impossible to measure the levels of signal all the time. We have to take snapshots of signals at various time.
- Oscilloscope shows graph of signal voltages by taking snapshots of the original signal in rapid repeated fashion. This process is called sampling.



Sampling Control

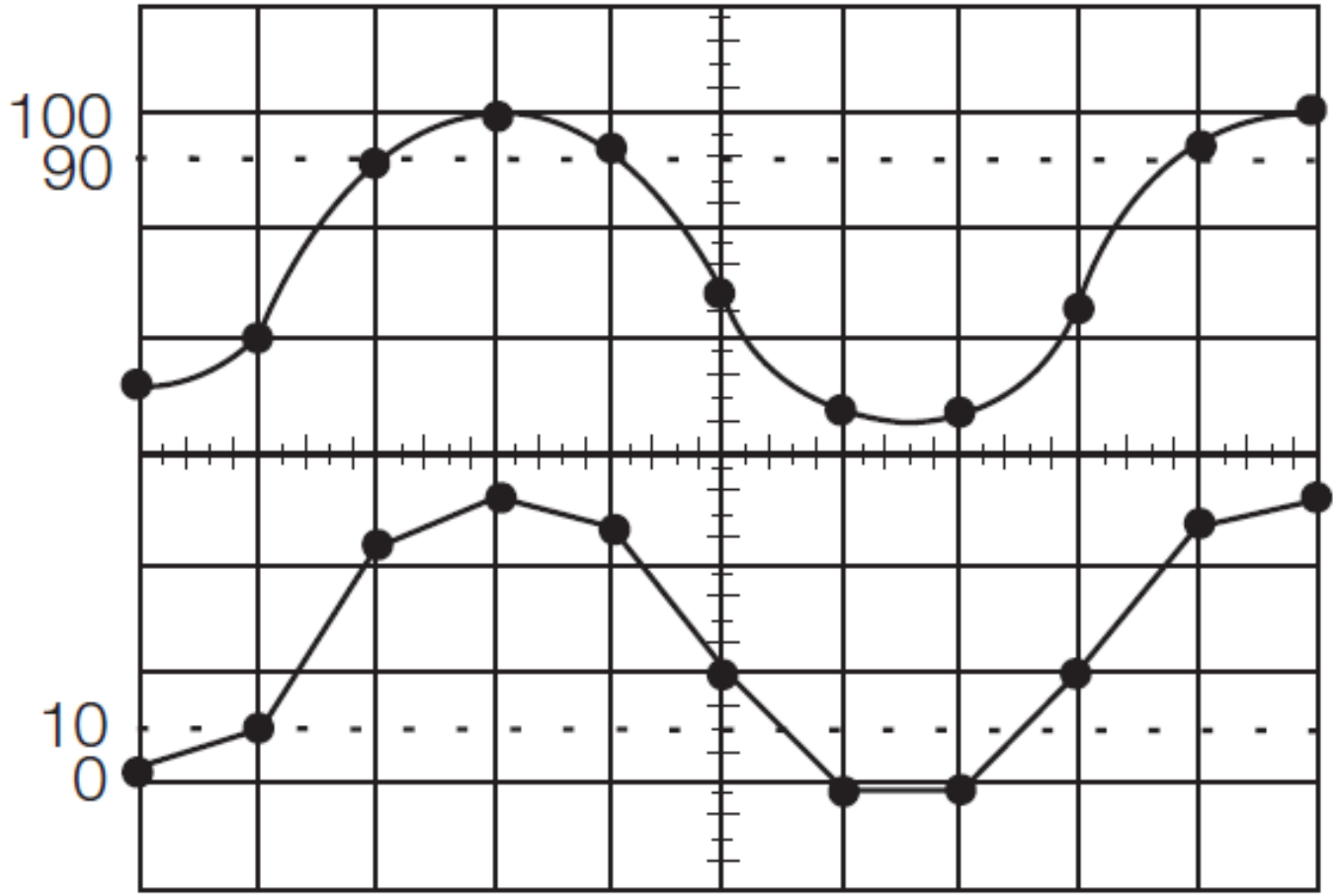
- In order to accurately measure the signal waveform, we must sample the signal at adequate frequency.



Reproduction of Signal

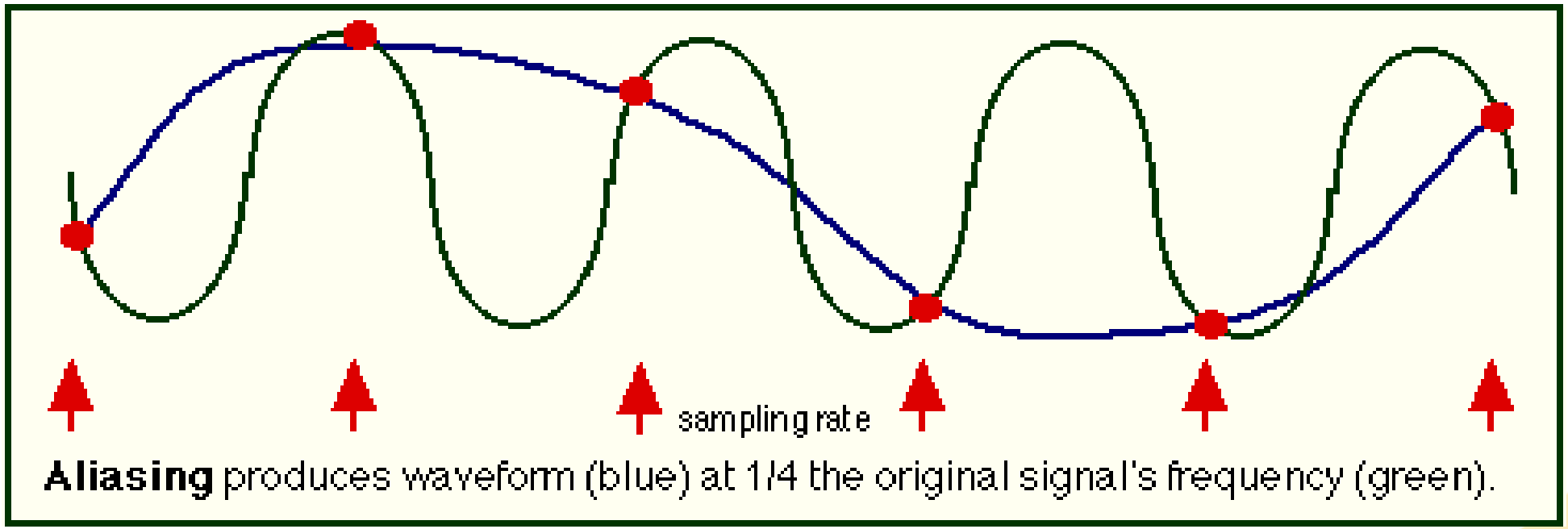
Sine Wave Reproduced using Sine x/x Interpolation

Sine Wave Reproduced using Linear Interpolation



Aliasing Effect

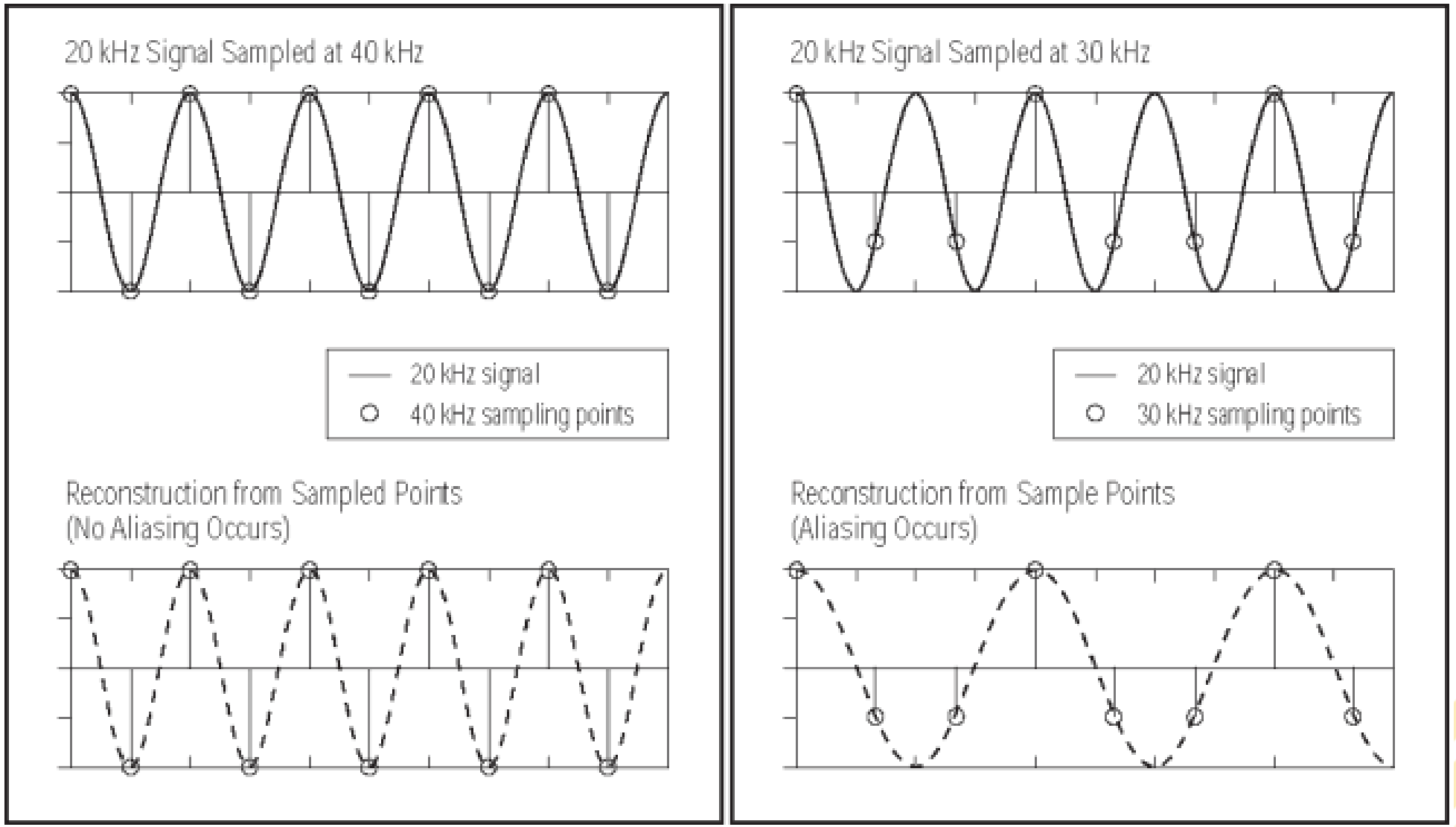
- If the sample rate isn't fast enough, high-frequency components can "fold down" into a lower frequency, causing aliasing in the display.
- Example: **Undersampling** of a 100 MHz sine wave introduces aliasing effects.



Aliasing produces waveform (blue) at 1/4 the original signal's frequency (green).

Sampling in Different Frequencies

Undersampling





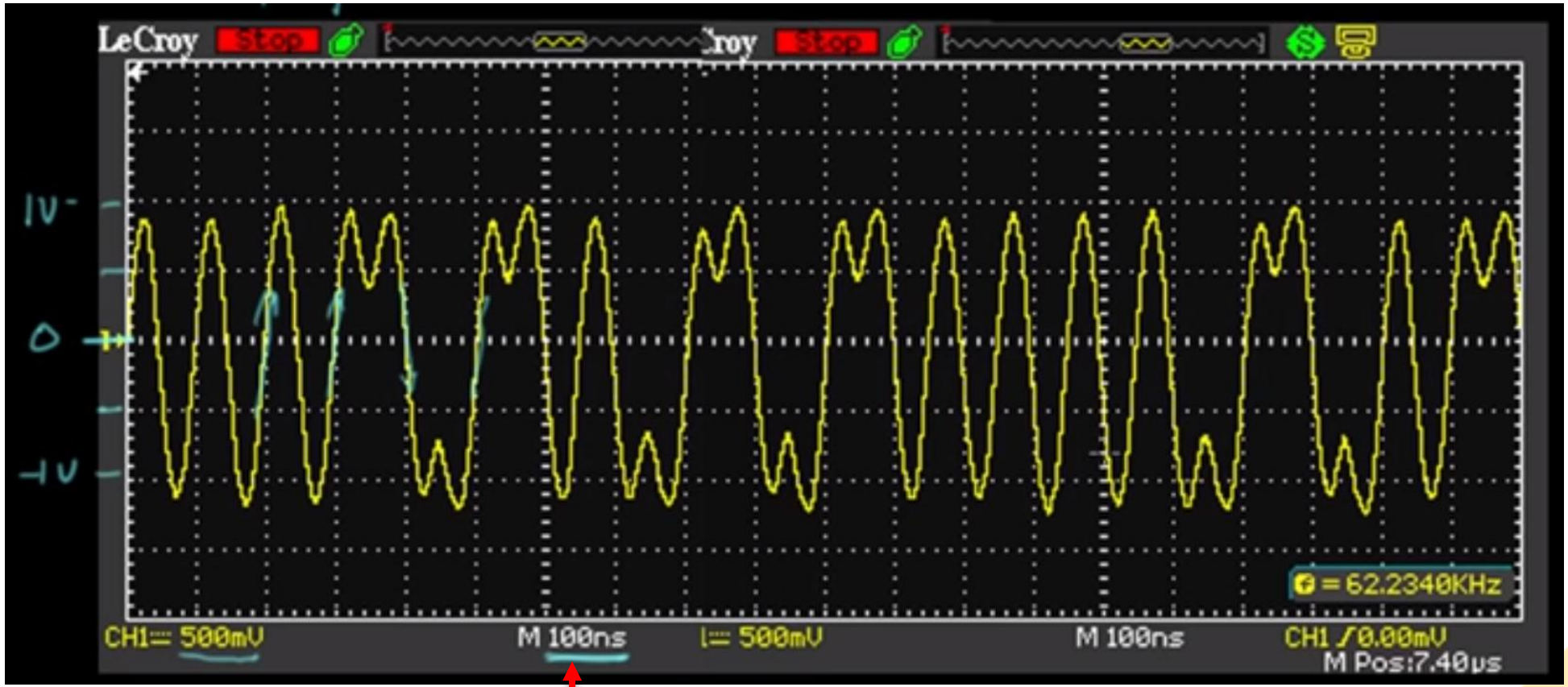
Sampling Rate Requirements

- In order to accurately reconstruct a signal and avoid aliasing, Nyquist theorem says that the signal must be sampled **at least twice** as fast as its **highest frequency** component.
  - i.e. If the highest frequency == 10 Hz, then the sampling must be at least 20 Hz!

- More example: a signal has a frequency range from 10 Hz to 20 KHz.
- Then the sampling rate should be 2 * 20 KHz = 40 KHz

Physical Transmission

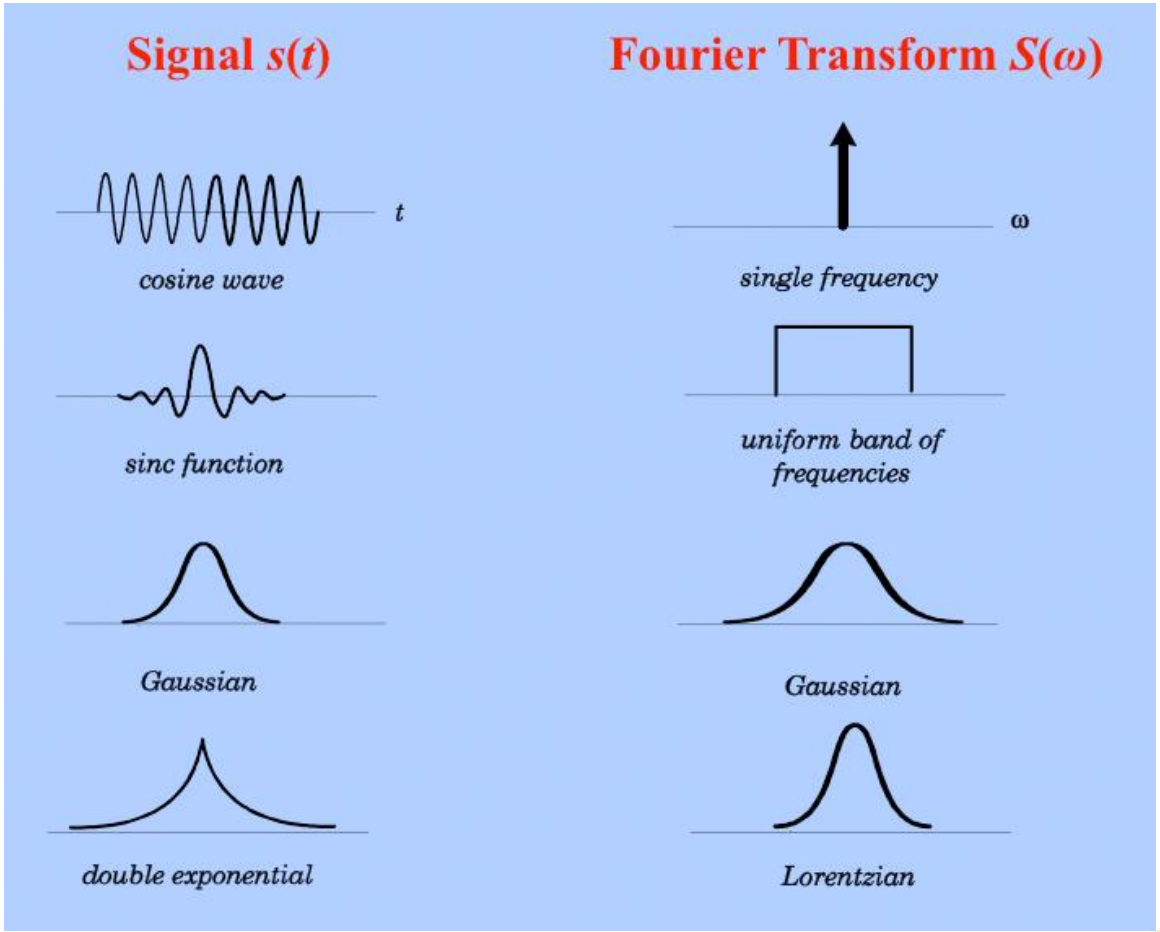
- Example: The signal detected is as shown in a **time-domain graph** below.



Frequency Domain Transformation

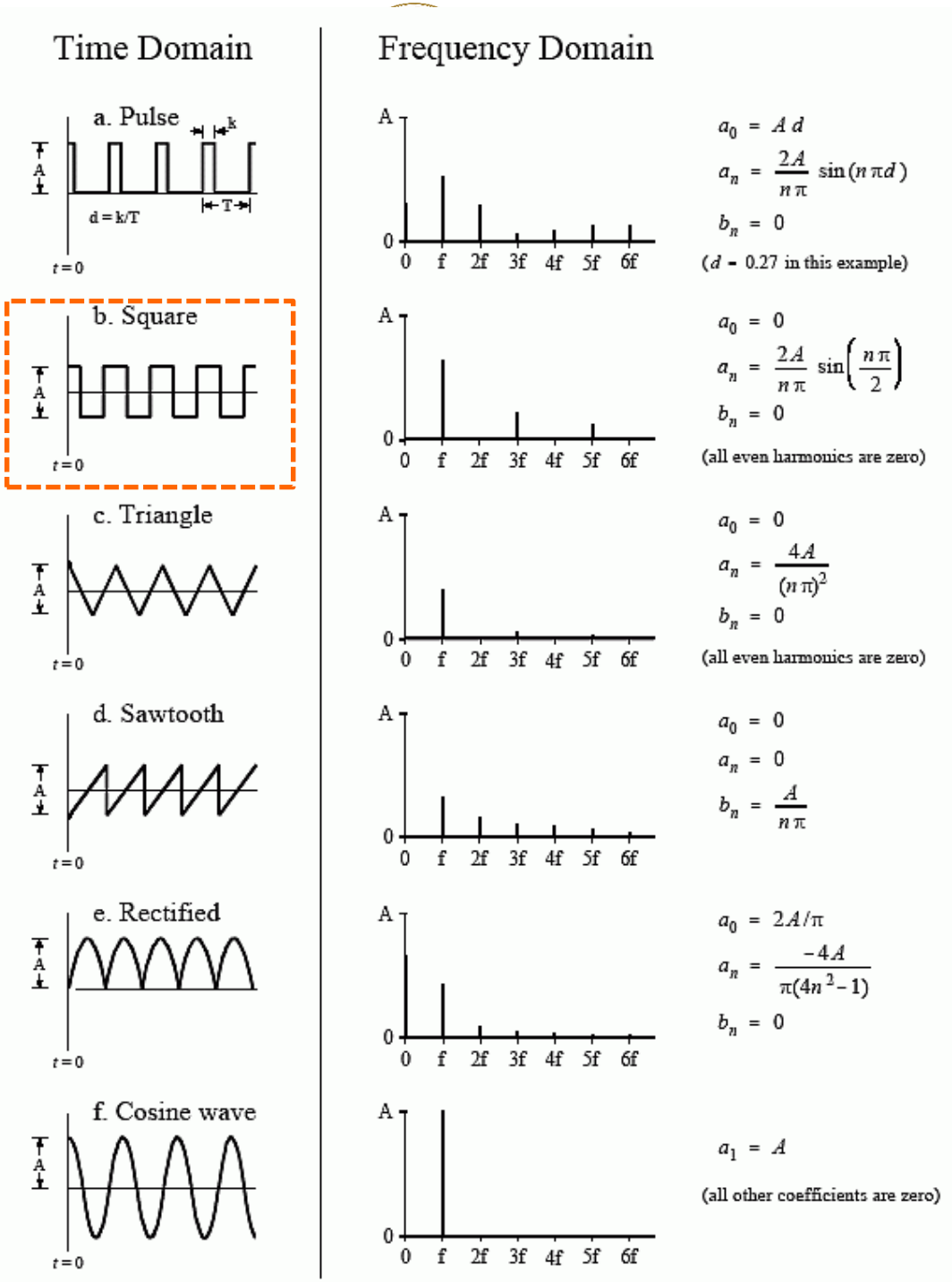
- Sometimes, we don't want to draw a lot to describe signals. So, we change our representation of waveforms to representation of waveforms in terms of sine waves where a sine wave of a particular frequency staring at original can be represented by a vertical line in **frequency domain** plot.

Time domain



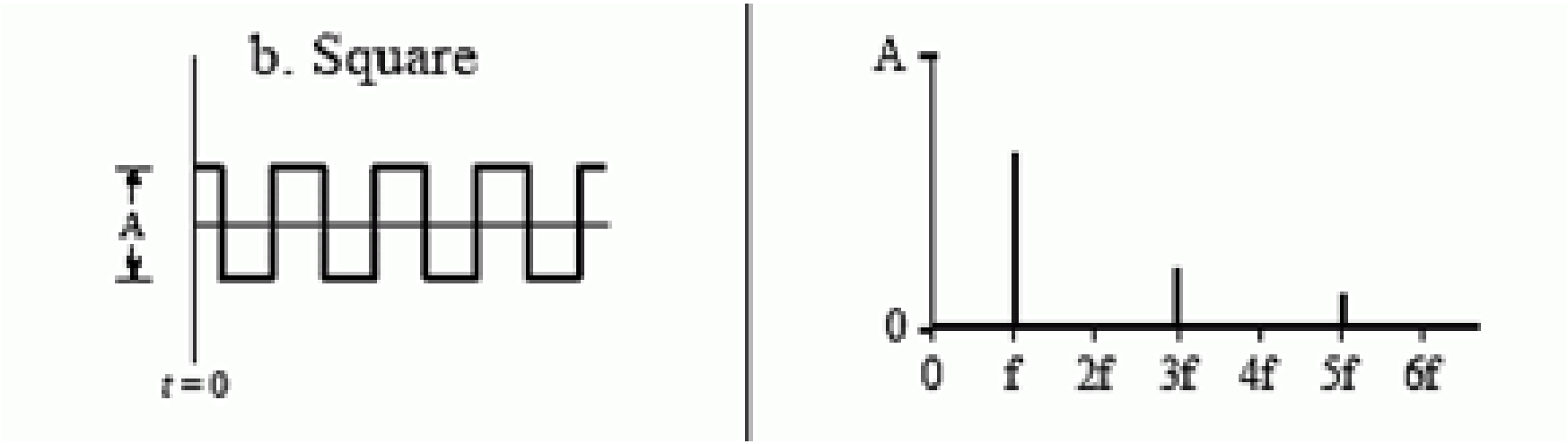
Frequency domain

Fourier Transform



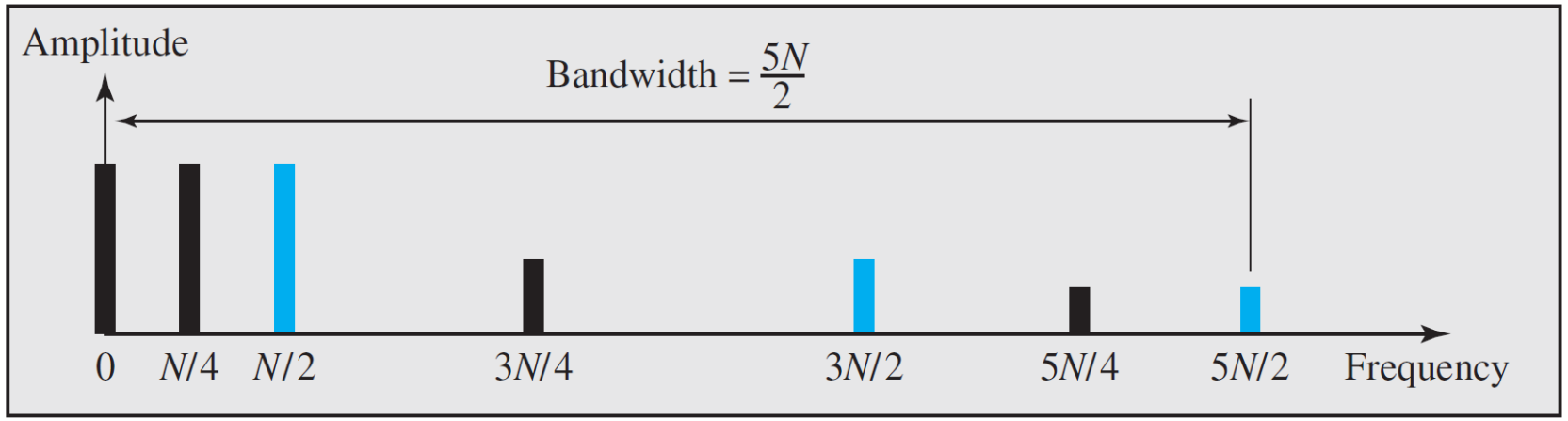
Square wave signal

- A digital signal (square wave) is a composite signal with an infinite bandwidth.

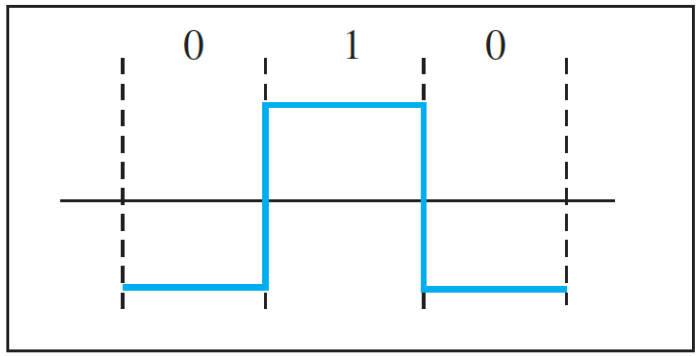


Odd harmonics

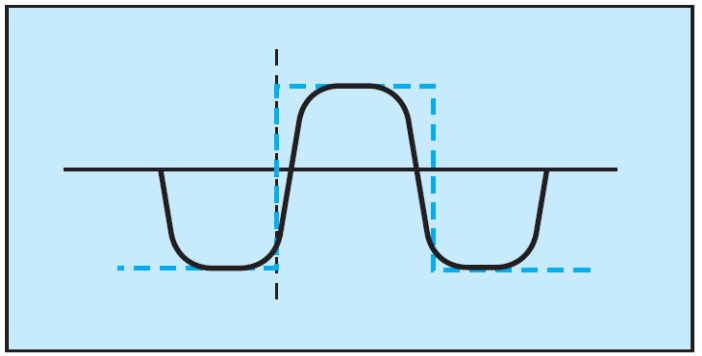
Digital Signals

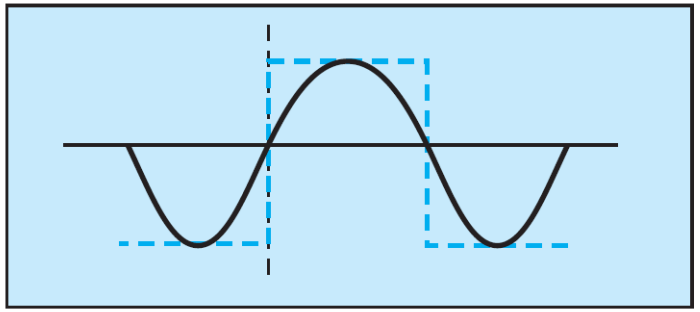


Digital: bit rate N

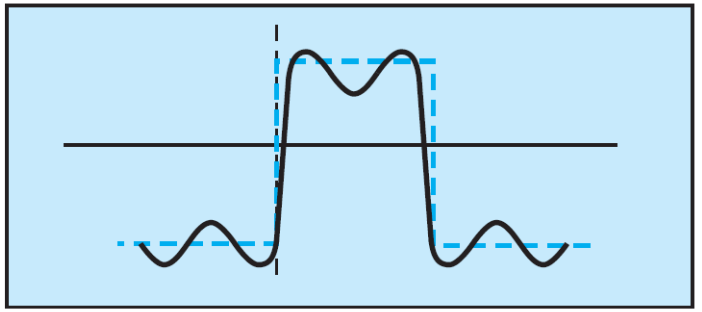


Analog: $f = N/2$ and $3N/2$





Analog: $f = N/2$



Analog: $f = N/2$, $3N/2$, and $5N/2$

Noise

Data → Square wave

Noise

Noise+Data

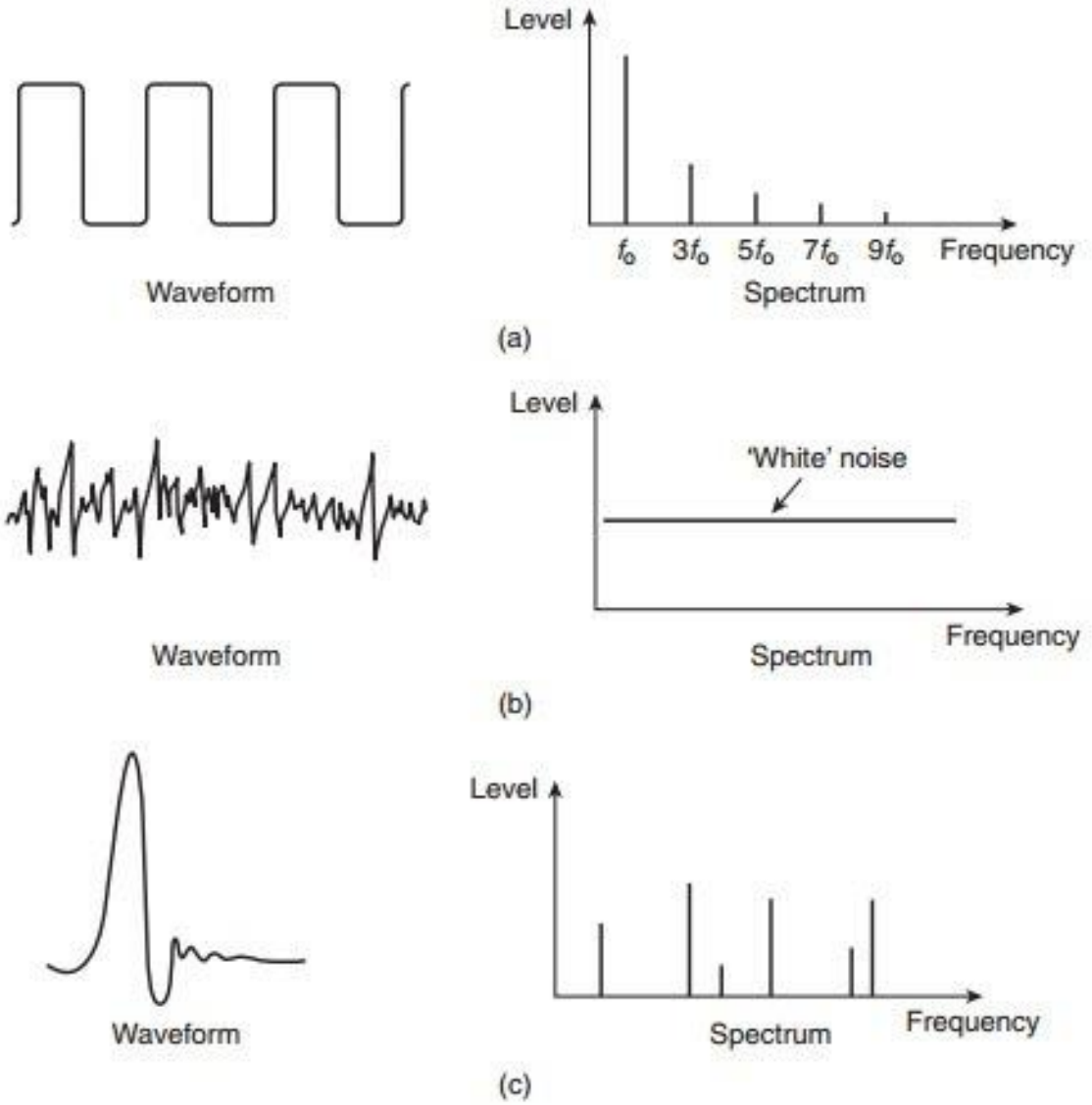
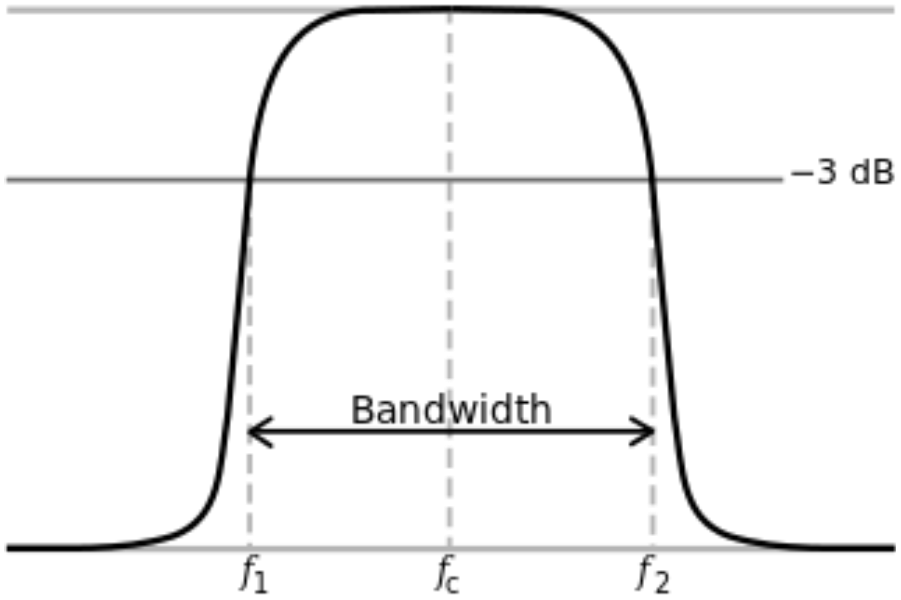


Figure 1.3: (a) A periodic signal repeats after a fixed time and has a simple spectrum consisting of fundamental plus harmonics. (b) An aperiodic signal such as noise does not repeat and has a continuous spectrum. (c) A transient contains an anharmonic spectrum.

Bandwidth

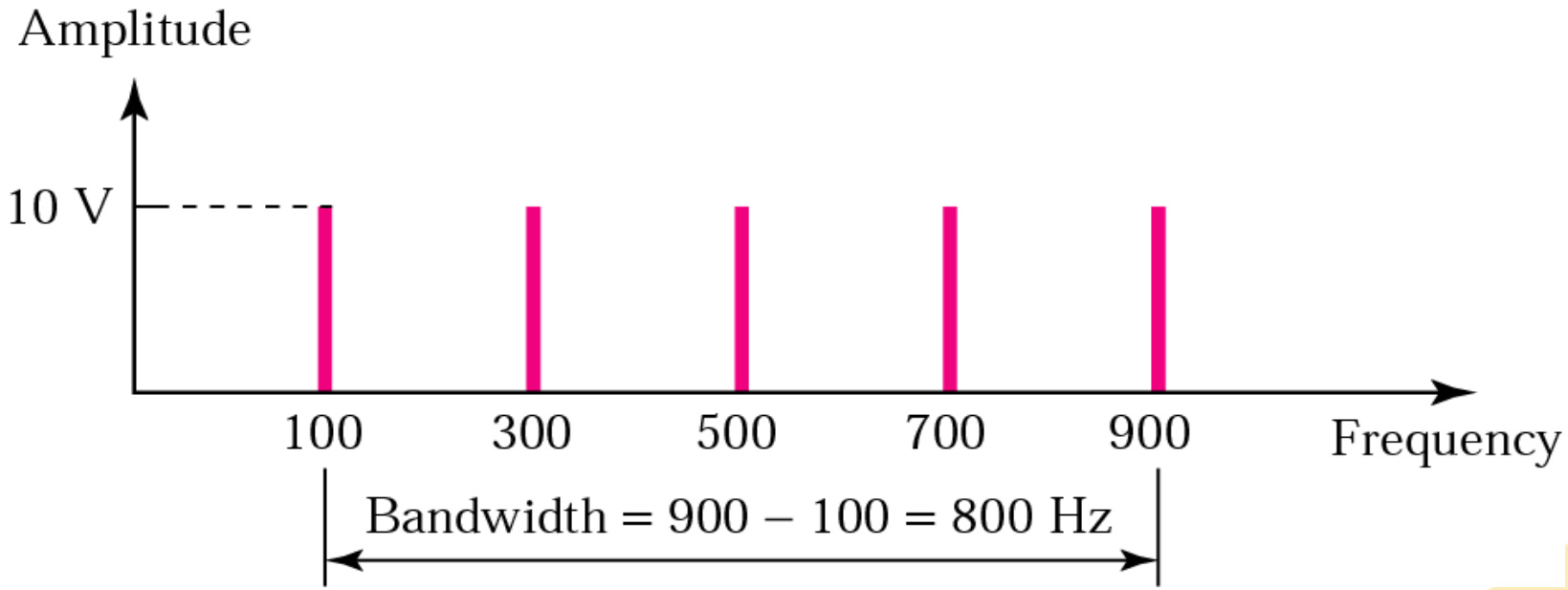
- Bandwidth is the range of frequencies that a medium can allow signals to pass through.
- It is **the difference between the highest and the lowest frequencies** that the medium can satisfactorily pass
- Some oscilloscopes measure signals and report the signals in frequency domain.



Example

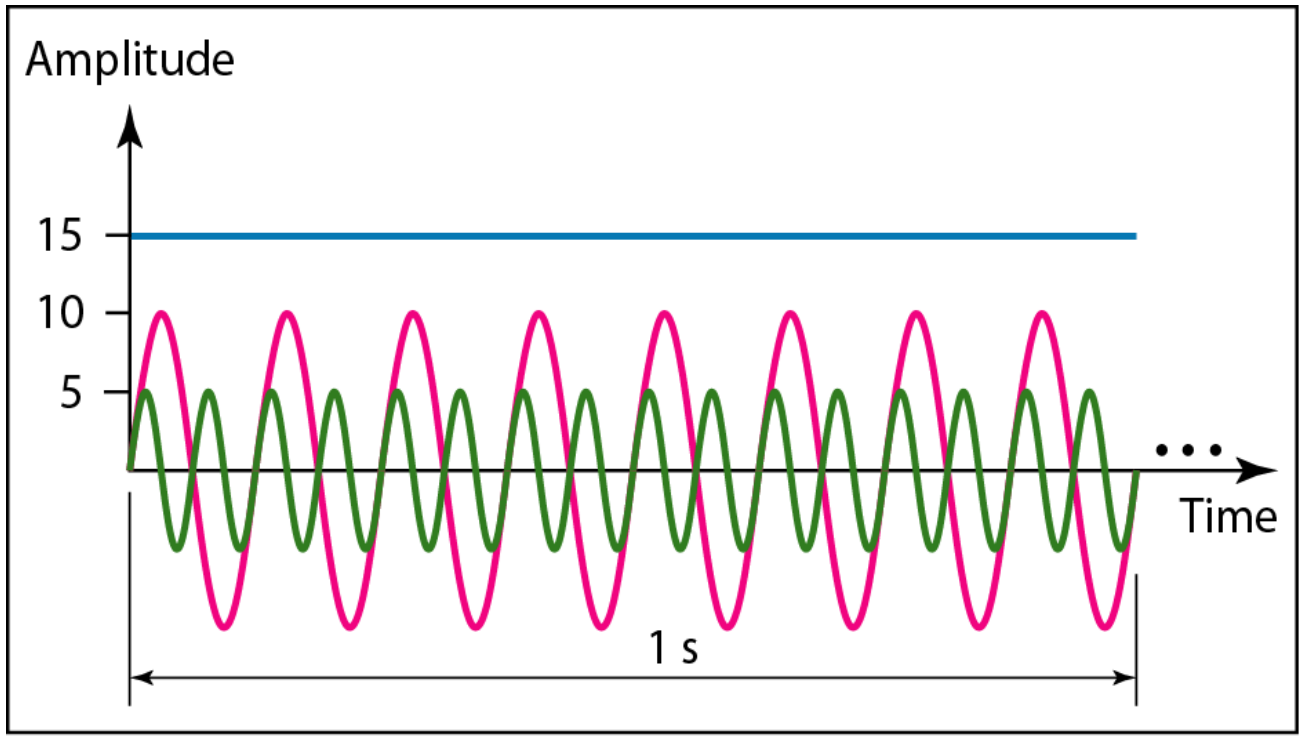
- If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, 900 Hz, what is the bandwidth ? Draw the spectrum, assuming all components have a maximum amplitude of 10V.

$$B = f_{high} - f_{low} = 900 - 100 = 800\ Hz$$

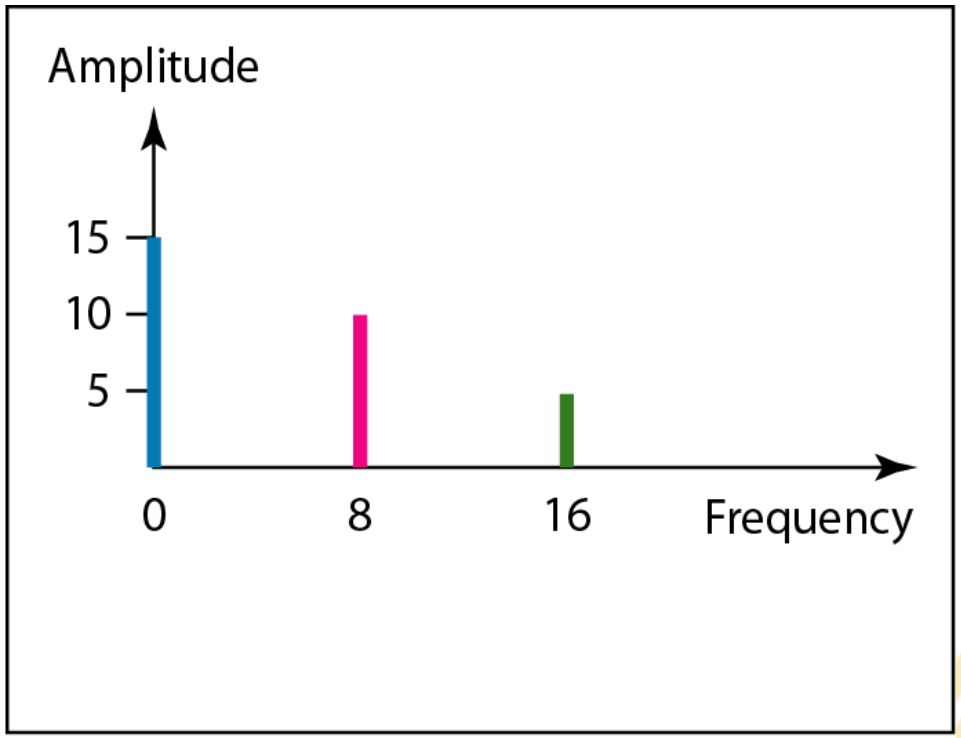


Example

- Draw the frequency domain of these signals



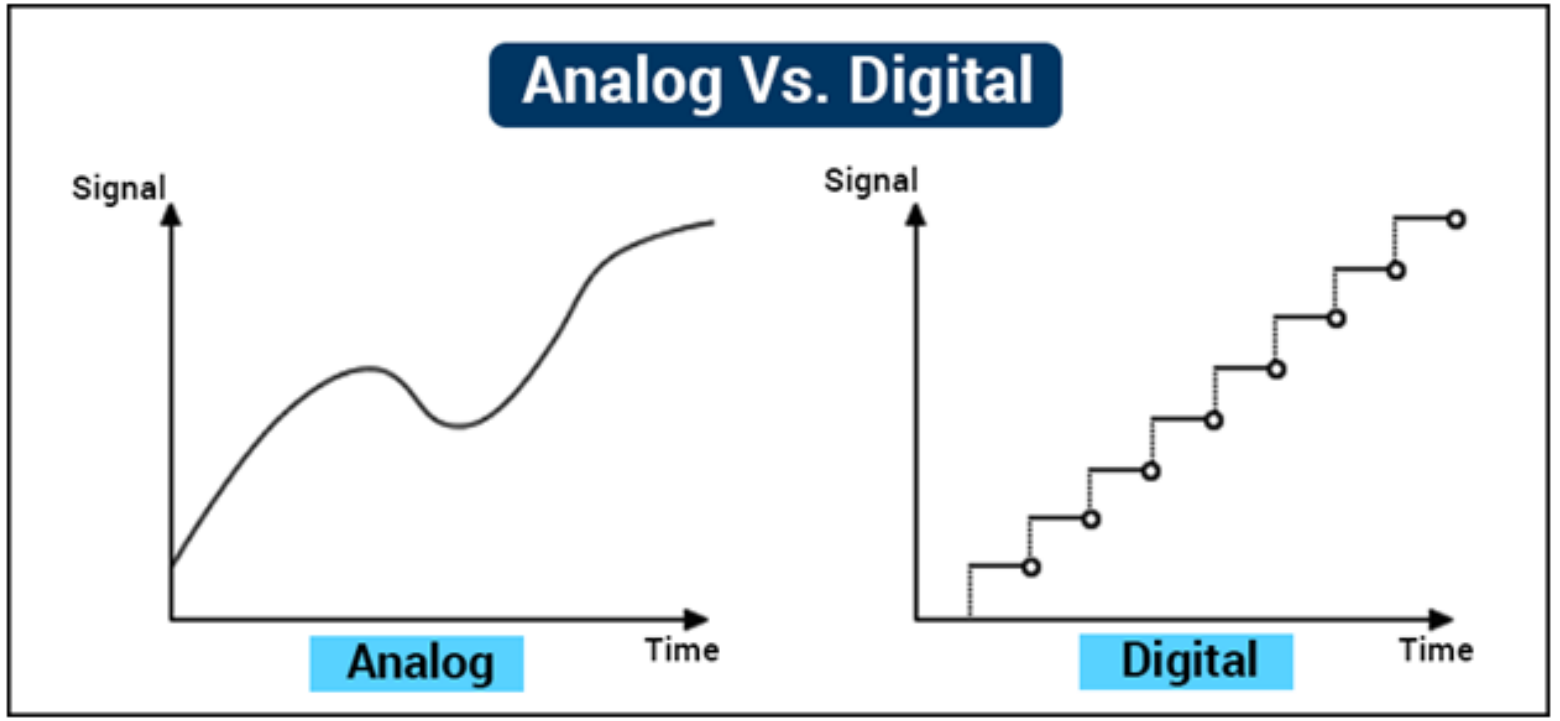
a. Time-domain representation of three sine waves with frequencies 0, 8, and 16



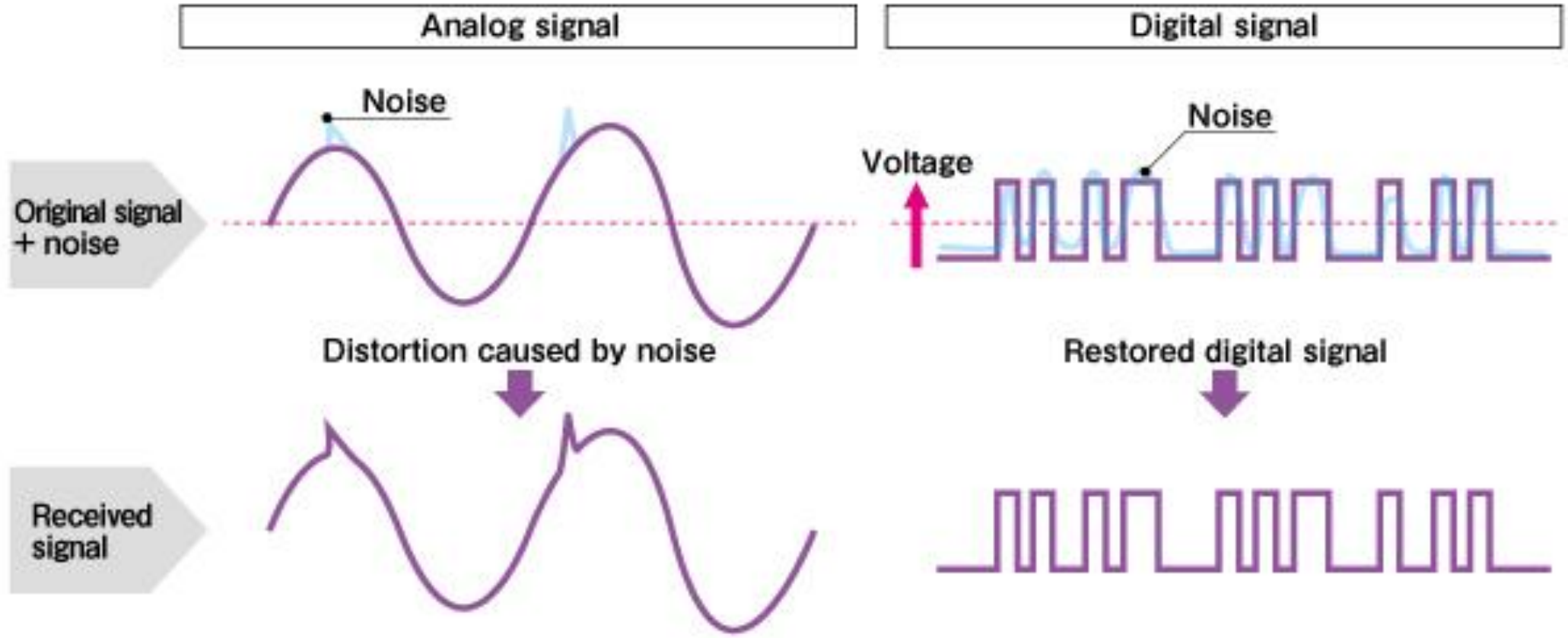
b. Frequency-domain representation of the same three signals

Analog vs. Digital Signals

- Analog: information can take any value
- Digital: information can only take on a certain set of values.



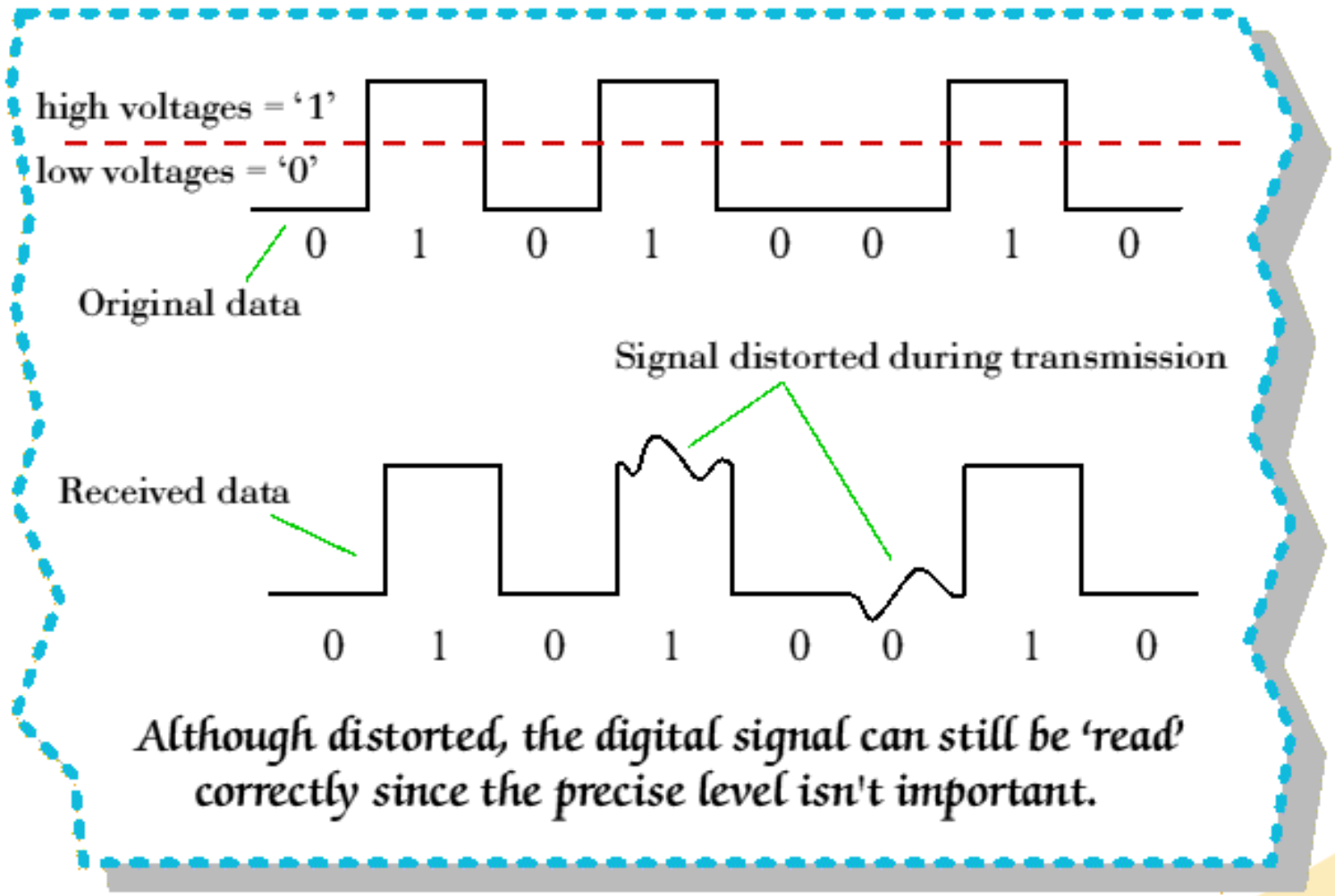
Why Digital Signal?



Digital signal is easier to read easily and accurately.
Digital signal can tolerate noise very well.

Why Digital Signals?

- Digital signals can tolerate noise more than analog signals.

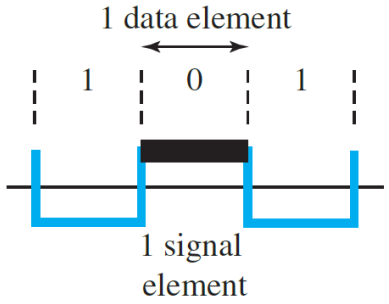


Data Transmission

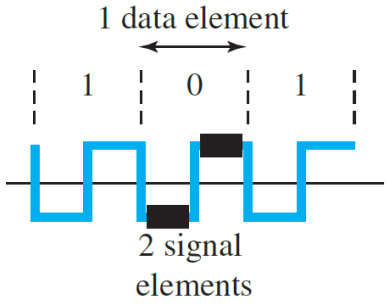
- In communication, our goal is to send information.
- The information is usually represented as data elements.
- When sending data to the recipient, the data have to be converted into signals so that they can travel through the medium (channels).
- Data elements are what we need to send. Signal elements are what we can send.
- The amount of data that can be sent over a unit time is called **data rate**.
- The amount of signals that can be sent over a unit time is called **signal rate**.

Digital Data to Digital Signal Representation

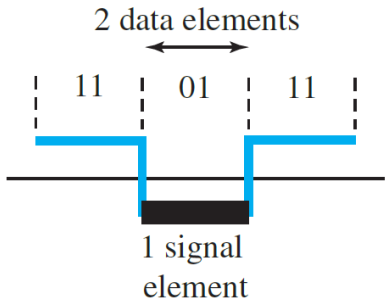
- Digital data from a computer must be converted into digital signals in order to be transmitted over wires.
- One or a group of signal elements (symbols) can represent one or a group of data elements.



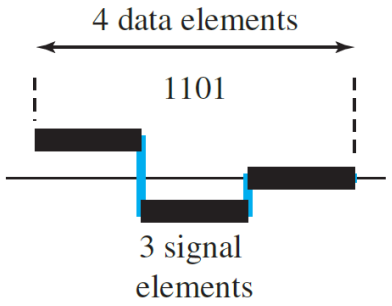
a. One data element per one signal element ($r = 1$)



b. One data element per two signal elements $\left(r = \frac{1}{2}\right)$



c. Two data elements per one signal element ($r = 2$)

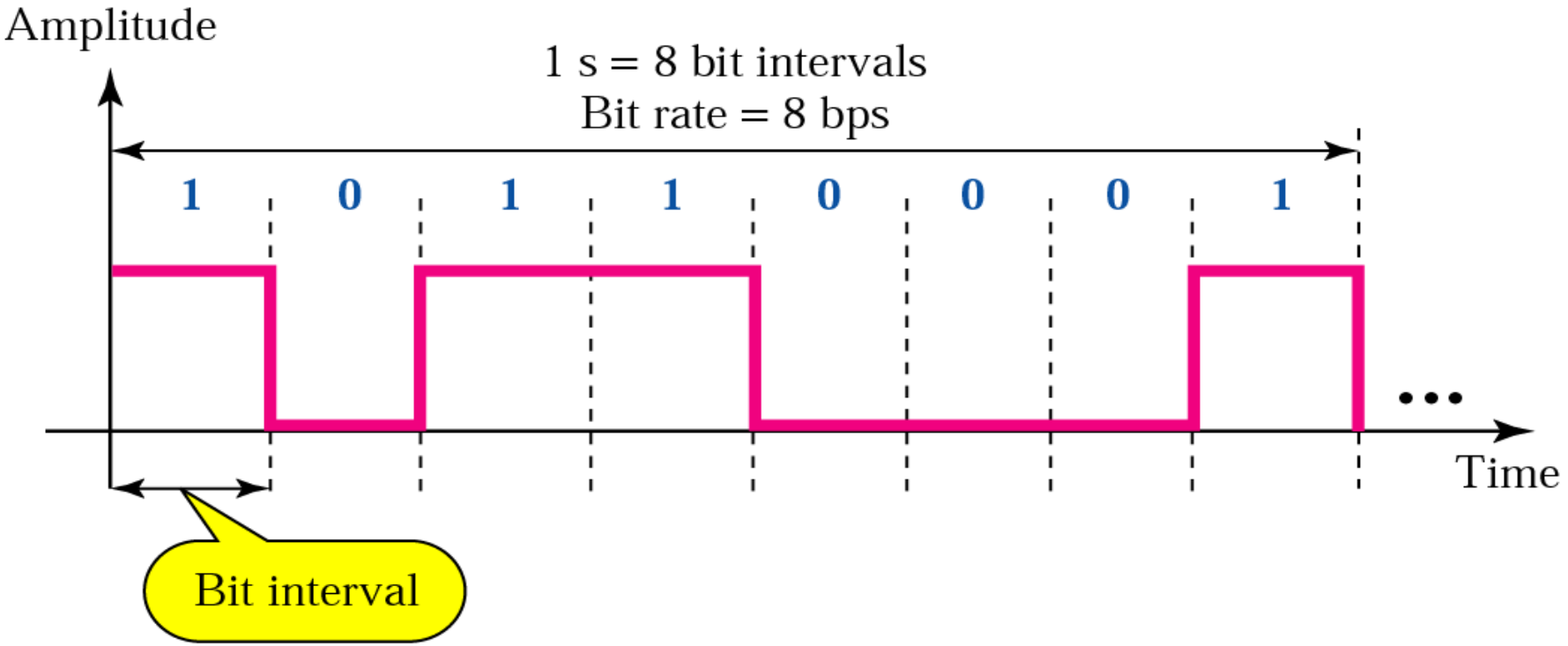


d. Four data elements per three signal elements $\left(r = \frac{4}{3}\right)$

r =
data element / signal element

Digital Signals

- Bit interval (bit duration) is the time required to send one single bit.
- Bit rate is the number of bit intervals per second (bits per second or bps)

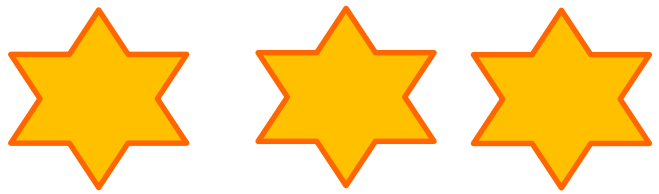


Example

- A digital signal has a bit rate of 2000 bps.
  What is the duration of each bit (bit interval) ?

- Answer:

  Note
  If bit rate = 1Mbps, then the bit interval = 1 μs

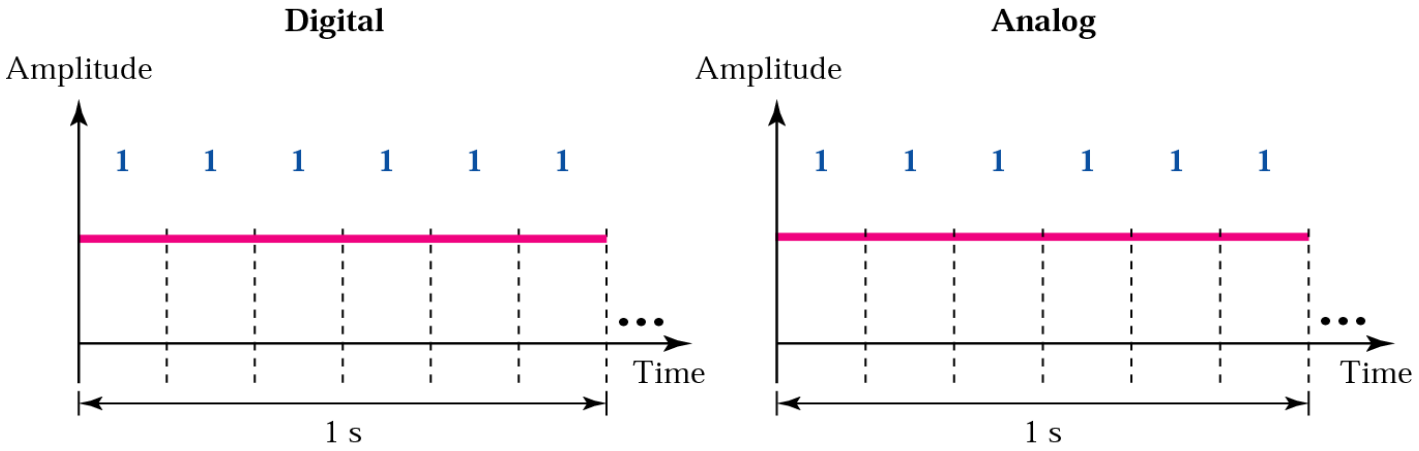


10^-3 → mili
10^-6 → micro
10^-9 → nano

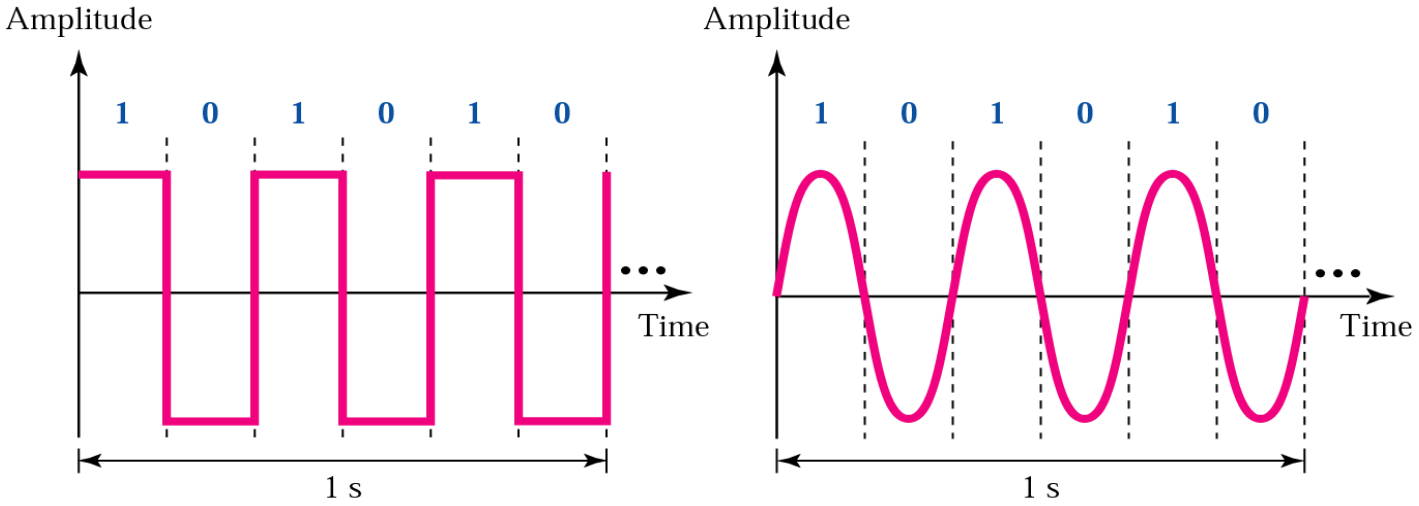
10^3 → kilo
10^6 → M
10^9 → G

Digital Data to Digital Signal Representation

- A digital signal is a composite signal with an infinite bandwidth



a. Best case, bit rate = 6, $f = 0$



b. Worst case, bit rate = 6, $f = 3$

Data Rate vs. Signal Rate

- **Data rate** defines the number of data elements (bits) sent in a unit time (1 second). The unit is often called bits per second (bps). The data rate is sometimes called bit rate. Usually the term data rate is used in connection to digital data transmission.

- **Signal rate** is the number of signal elements sent in a unit time (1 second). The unit is often called baud (signal elements per second). The signal rate is sometimes called modulation rate or baud rate.

- Our goal in data communication is to increase the data rate while decreasing the signal rate. Analogy: we need to carry more people in fewer vehicles to prevent traffic jams.
  - ✓ Increasing the data rate increases the speed of transmission.
  - ✓ Decreasing the signal rate decreases the bandwidth requirement.

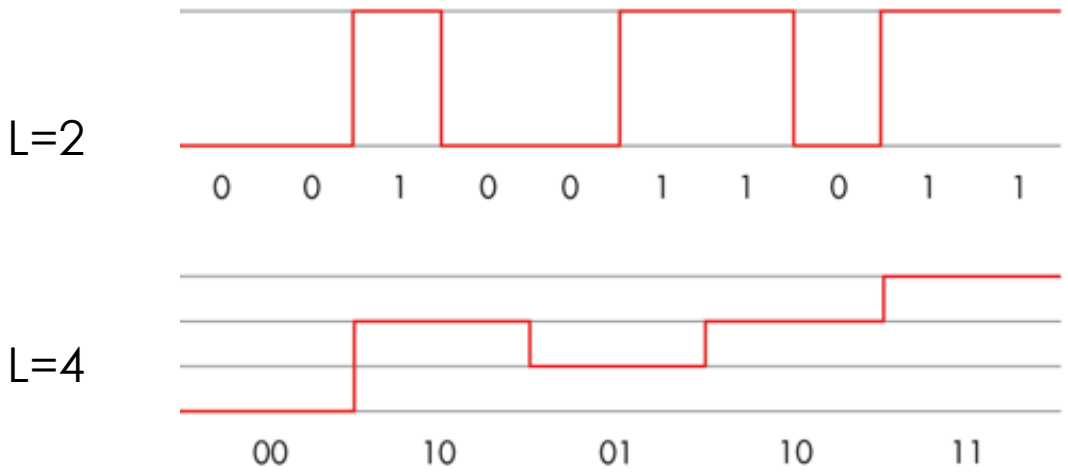
- In any transportation system, we can only support a limited number of signal elements at a time. This is sometimes called bandwidth.

Digital Data to Digital Signal Representation

- The more level, the more bits you can send in the same period of time.



L = number of levels



- If L = 2, 1 symbol carries 1 bit (0 or 1) => r = 1
- If L = 4, 1 symbol carries 2 bits (00, 01, 10, 11) => r = 2
- If L = 8, 1 symbol carries 3 bits (000, 001, 010, …, 110, 111) => r = 3

$$r = \text{bit-symbol ratio} = \frac{\#\ \text{bits}}{\text{symbol}} = \log_2 L$$

Signal Rate (Baud Rate) and Bit Rate (Data Rate)

- Signal rate (S) = number of signal symbols transmitted in 1 second
- Bit rate (N) = number of bits transmitted in 1 second

- Given the same signal rate (e.g., 1 symbol per second)
  - ✓ If L = 2, we can transmit 1 bit per second (1 symbol represents 1 bit)
  - ✓ If L = 4, we can transmit 2 bits per second (1 symbol represents 2 bits)
  - ✓ If L = 8, we can transmit 3 bits per second (1 symbol represents 3 bits)
  - ✓ If L = 16, we can transmit 4 bits per second (1 symbol represents 4 bits)

4 = 2^2

8 = 2^3

16 = 2^4

- The relationship between data rate (N) and signal rate (S) is as follows

$$S = \frac{N}{r}$$

r = how many bits per symbol

- Since r depends on the data-symbol representation pattern, we can have
  - ✓ the best case when r is really large (difficult to do) and
  - ✓ the worst case when an infinite number of signals are used to represent only one bit.
  - ✓ Therefore, on average, we have

$$S_{avg} = c \times \frac{N}{r}$$

where c is an average case factor (usually c = 0.5)

Example 4.1

- A signal is carrying data in which one data element is encoded as one signal element (r = 1). If the bit rate (N) is 100 kbps, what is the average value of the baud rate if c is between 0 and 1?

- Solution:
  - ✓ If we assume that the value of average case factor (c) is 0.5, the average signal rate or baud rate is then

$$S_{avg} = c \times \frac{N}{r} = 0.5 \times \frac{100{,}000}{1} = 50{,}000 = 50\ kbaud\ /s$$



Digital Data to Signal Representation

- Remember that the bandwidth requirement for digital data to digital or analog signal conversion depends on
  - ✓ Bit rate
  - ✓ Quality of data to signal conversion

Let's do the exercise and have a Break !





Terminologies until this break

- Oscilloscope
- Aliasing Effect
- Sampling
- Minimum sampling rate
- Nyquist theorem
- Time-domain graph
- Frequency-domain graph
- Square-wave signal
- Bandwidth
- Data Rate & Signal Rate

Terminologies after the break

1. Low Pass Filter
2. Band-Pass Filter
3. Digital Transmission
4. Analog Transmission
5. Attenuation
6. **dB equation**
7. dBm equation



8. Distortion
9. Noise
10. **SNR equation**
11. Propagation Time
12. Throughput
13. Bandwidth-Delay Product
14. **Nyquist Bit Rate** **
15. **Shannon Bit Rate** **

Low Pass Filter

- Low Pass Filter is a device which only allows the low frequency components to pass through it and provides very high attenuation to the high frequency components (filter out high frequency components).

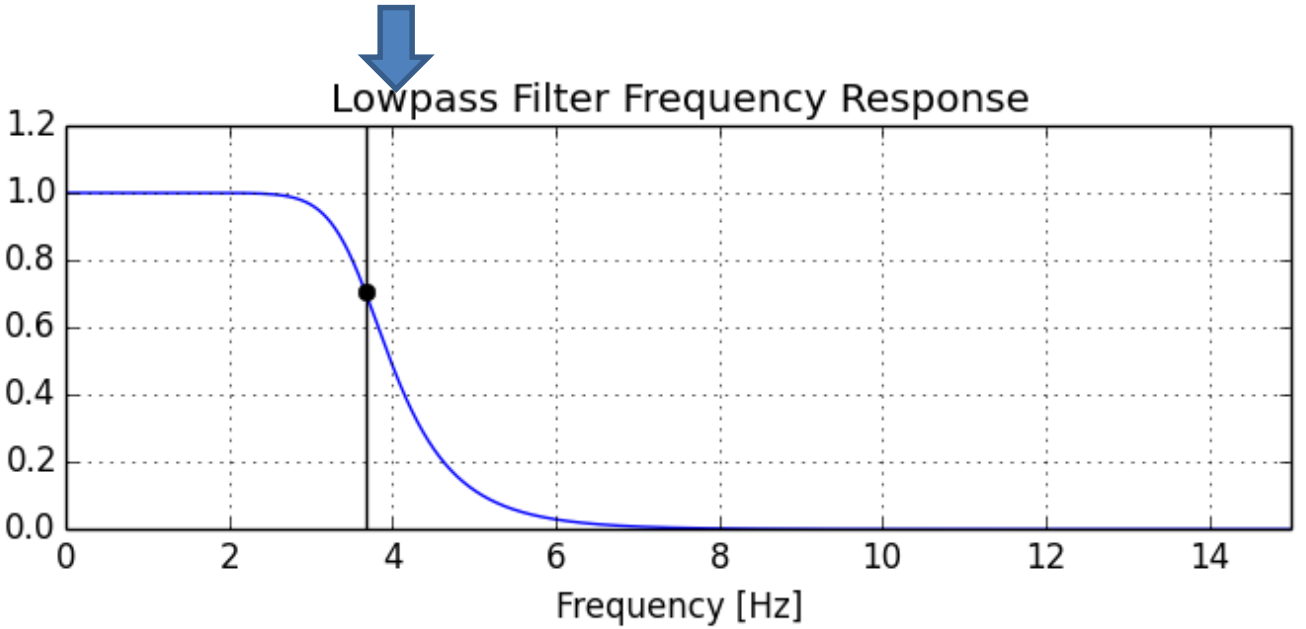
- Making electronic circuits for low frequency components or translating a higher frequency component into lower one makes our circuit simpler to implement

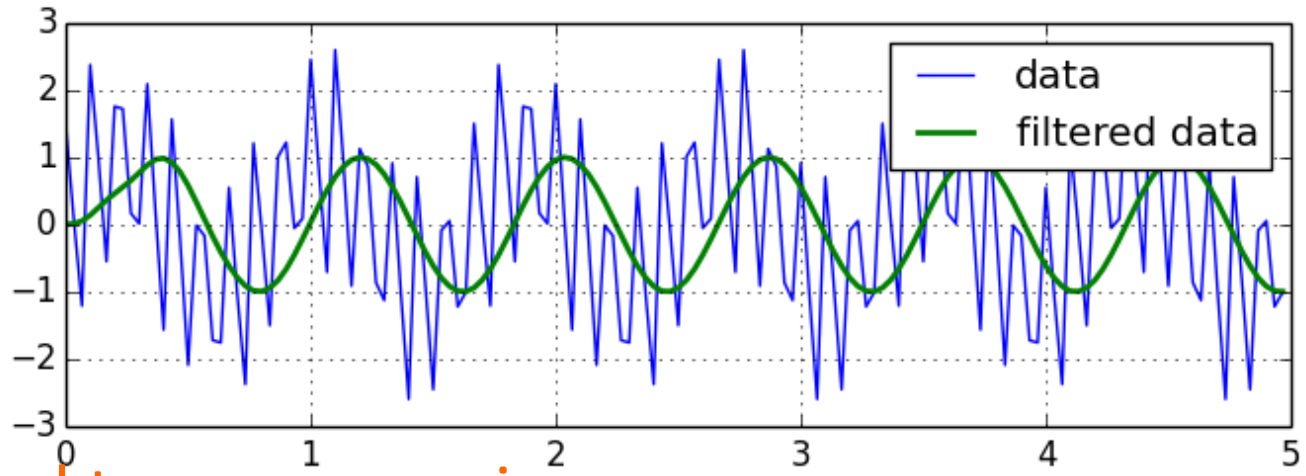
- Applications:
  - ✓ Audio Applications for Equalization purposes
  - ✓ Noise removal
  - ✓ Anti - Aliasing Filter - Major application for images and sounds.



Low Pass Filter

This low-pass filter will block frequency higher than this point (3.8 Hz?)



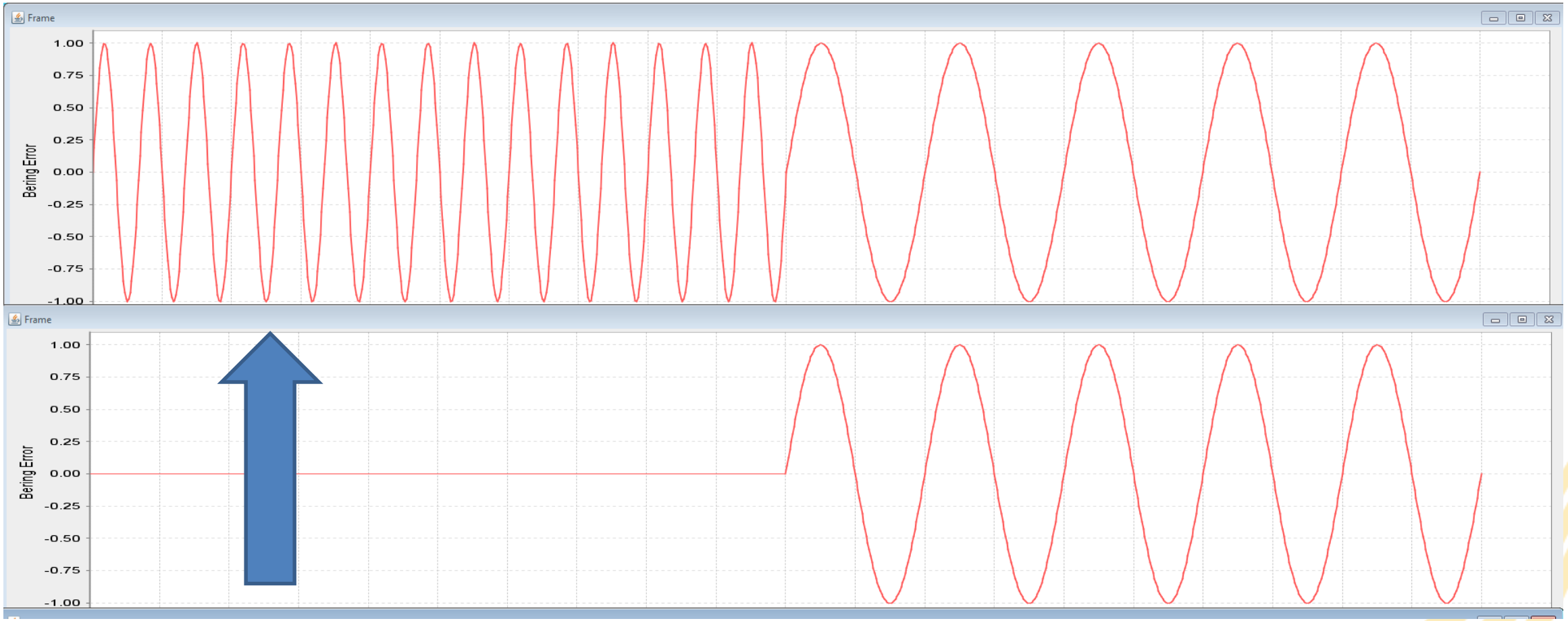


Low-pass filter is used to remove noise

Low Pass Filter

High frequency (many cycles)

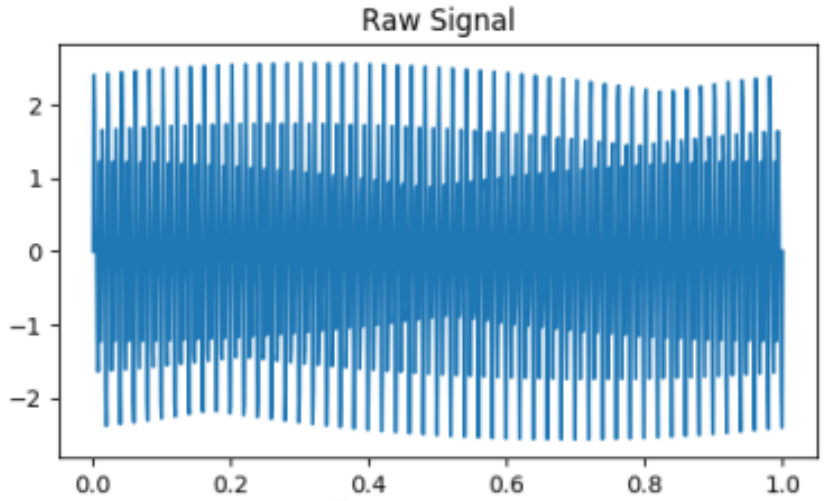
Low frequency (few cycles)



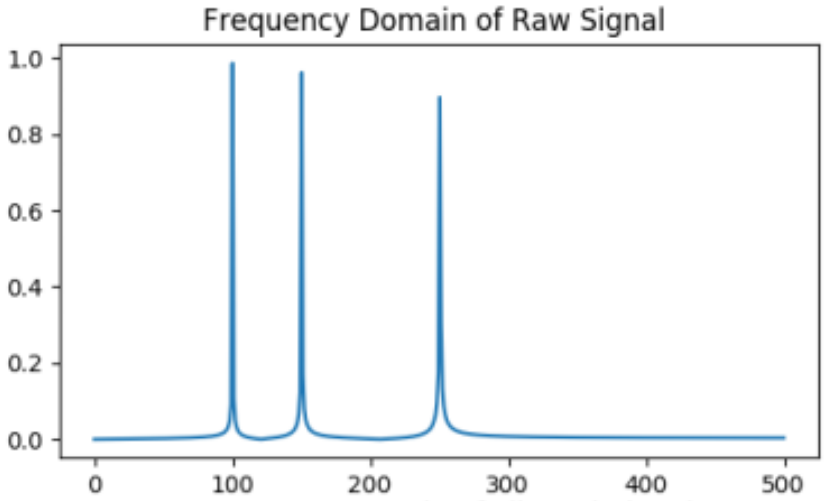
Low frequency will remain. High frequency will be removed

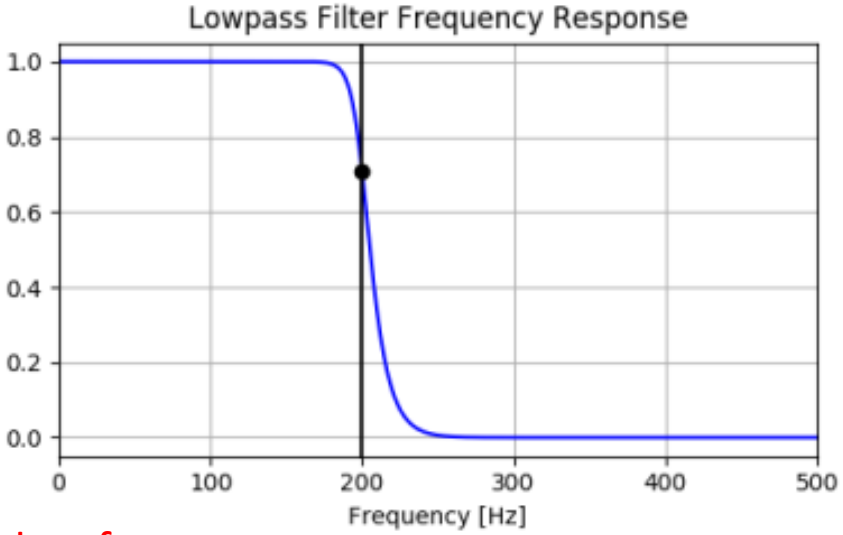
Low Pass Filter

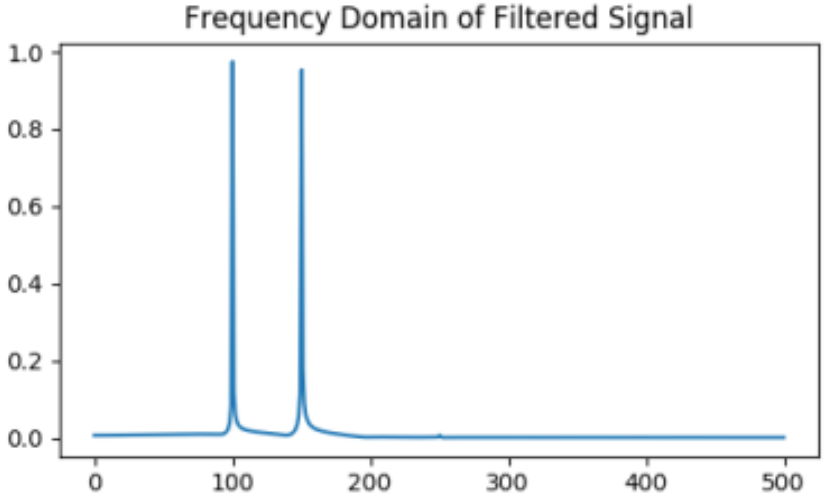
1. Time Domain



2. Original Frequency Domain







3. Lowpass Filter

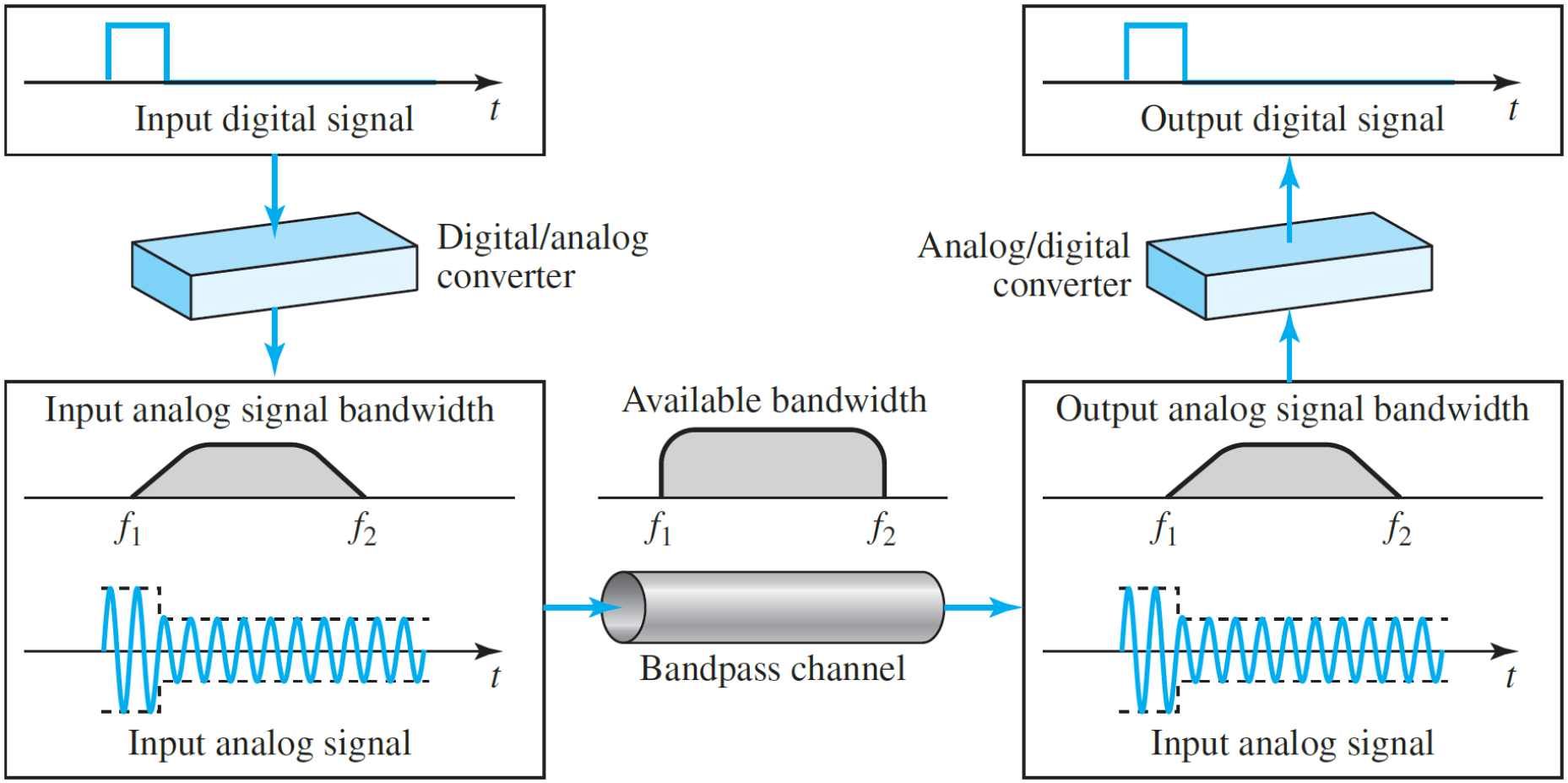
→ Will remain only low frequency

→ In this filter, it will block frequency higher than 200 Hz

4. Frequency Domain after passed the Low Pass Filter 200 Hz

Band-Pass Filter

Band-pass filter is used to select signals in the specific frequency.
Select only frequency in the band we want.

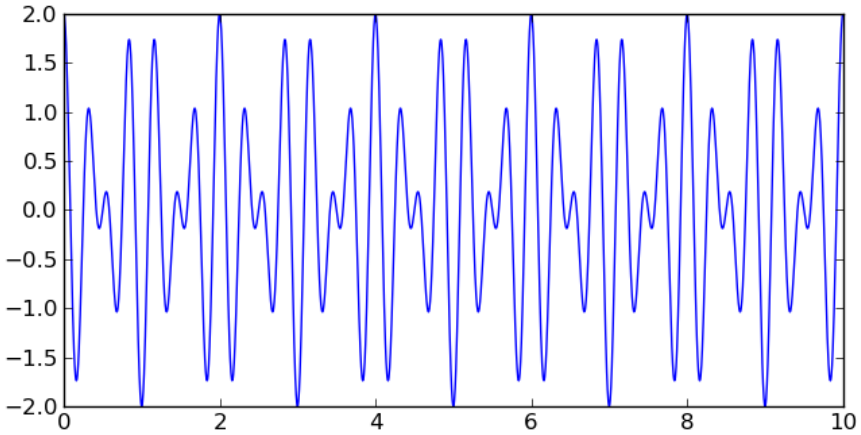


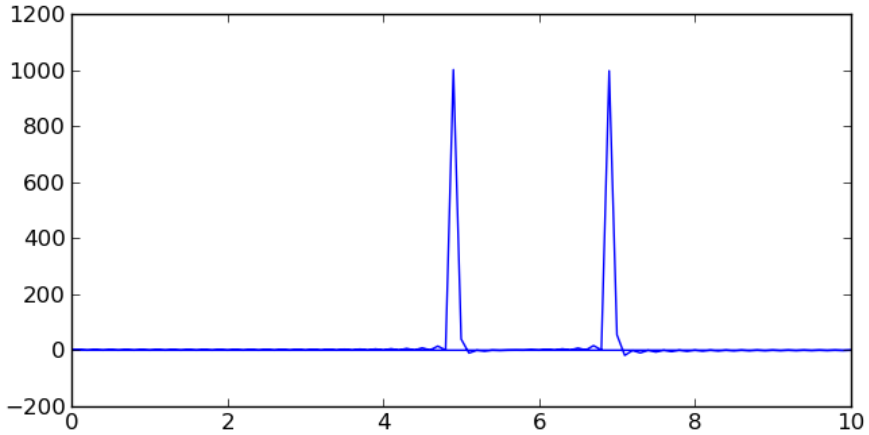
For example, FM radio frequency range

Band Pass Filter

There are signals with different frequency (multiplexed signals) in the air.

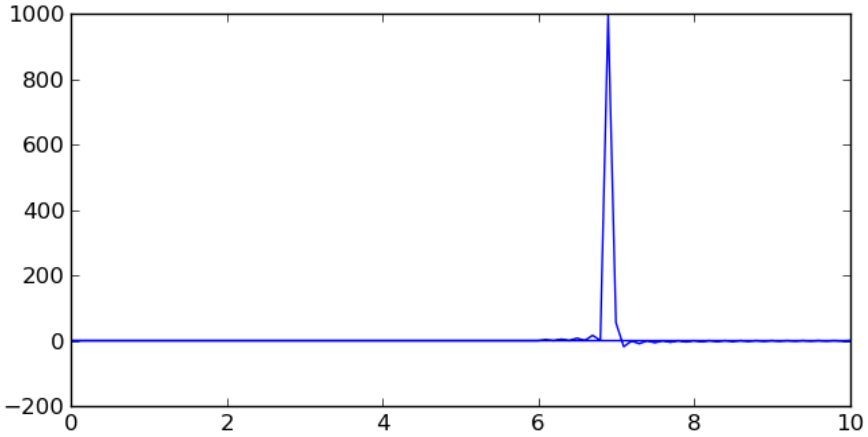
(1) time-domain

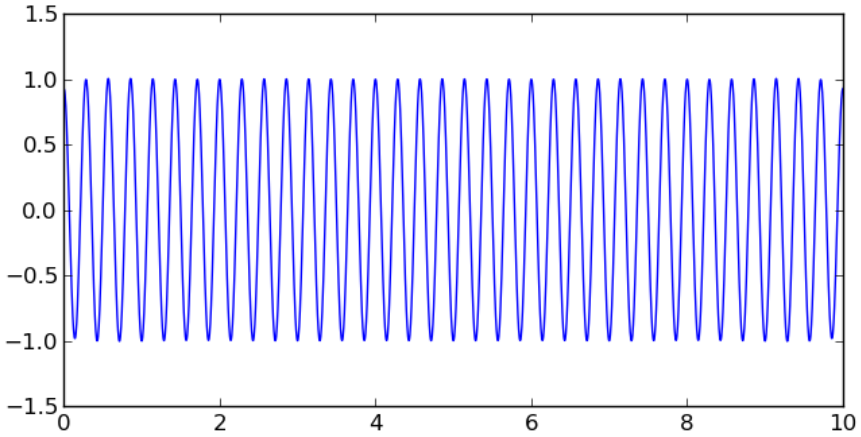




(2) frequency-domain

(3) frequency-domain



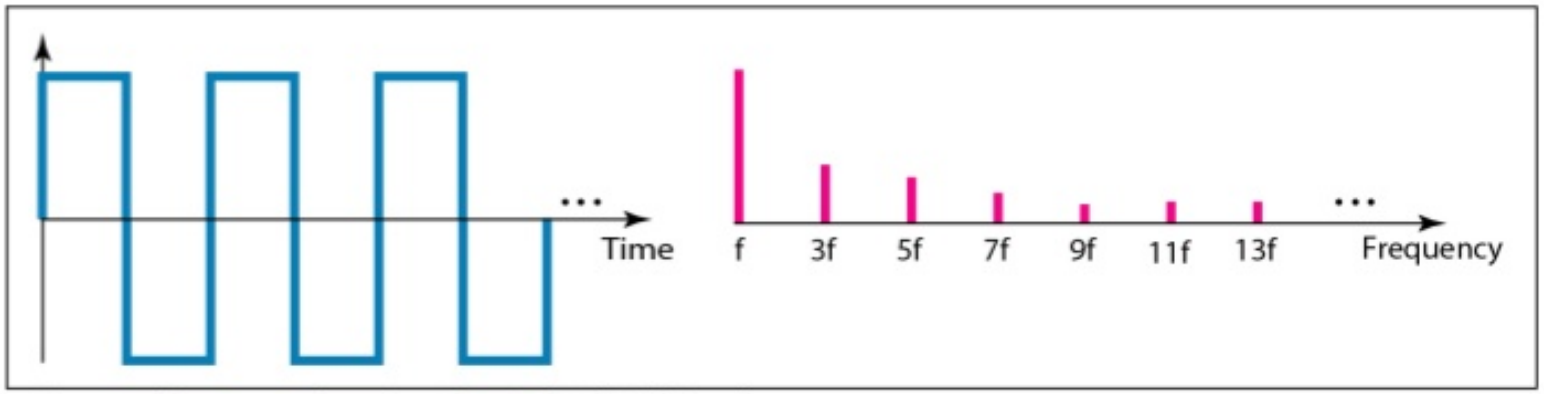


(4) time-domain

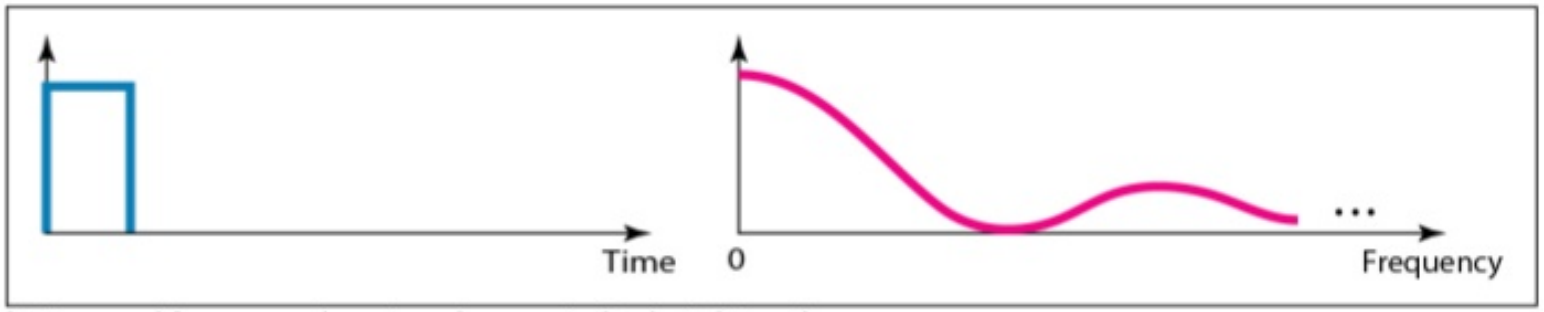
If we tune our radio to only channel (one frequency) --> use Band Pass filter

Digital Transmission

- **Digital signals require infinite frequency**.
- Therefore, the medium used for digital signal transmission must allow all infinite frequency to pass through. **The medium must be a low pass channel with cut-off point at high frequency**.



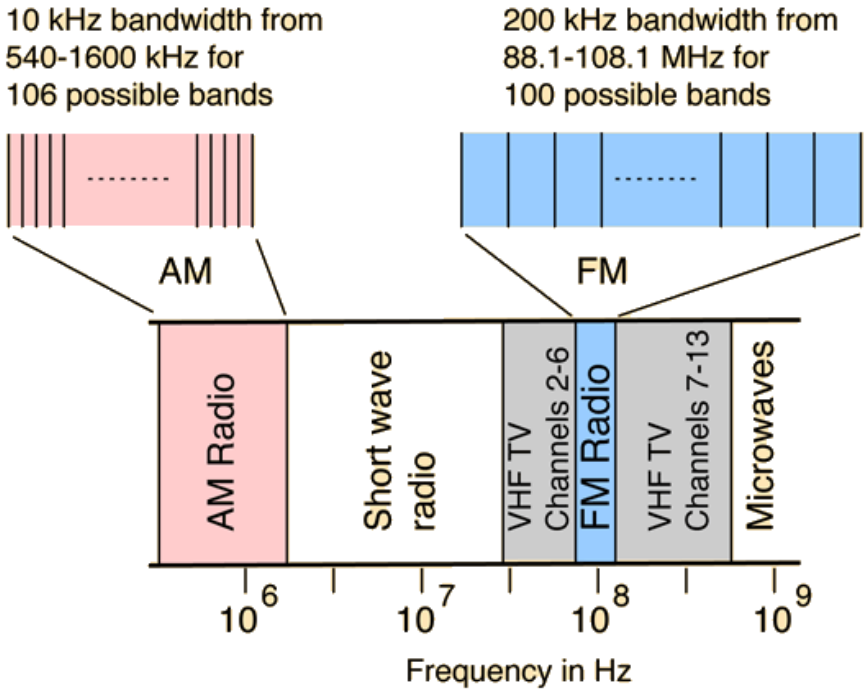
a. Time and frequency domains of periodic digital signal



b. Time and frequency domains of nonperiodic digital signal

Analog Transmission

- A analog signal normally has a ***narrower*** bandwidth than a digital signal with frequencies between f_1 and f_2
- Therefore, analog signals should require medium which supports **band-pass channel**.
- A signal of frequencies between f_1 and f_2 can be moved to frequency between f_3 and f_4 in order to pass through the band-pass channel. This concept is important because it is fundamentally used in frequency multiplexing scheme.



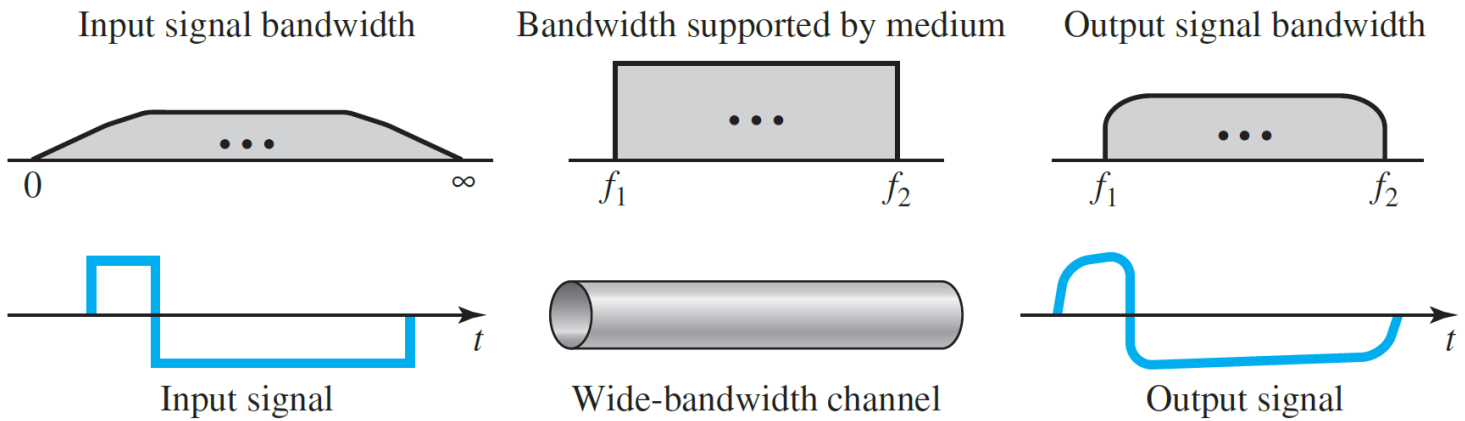


Radio

Data Transmission and Media

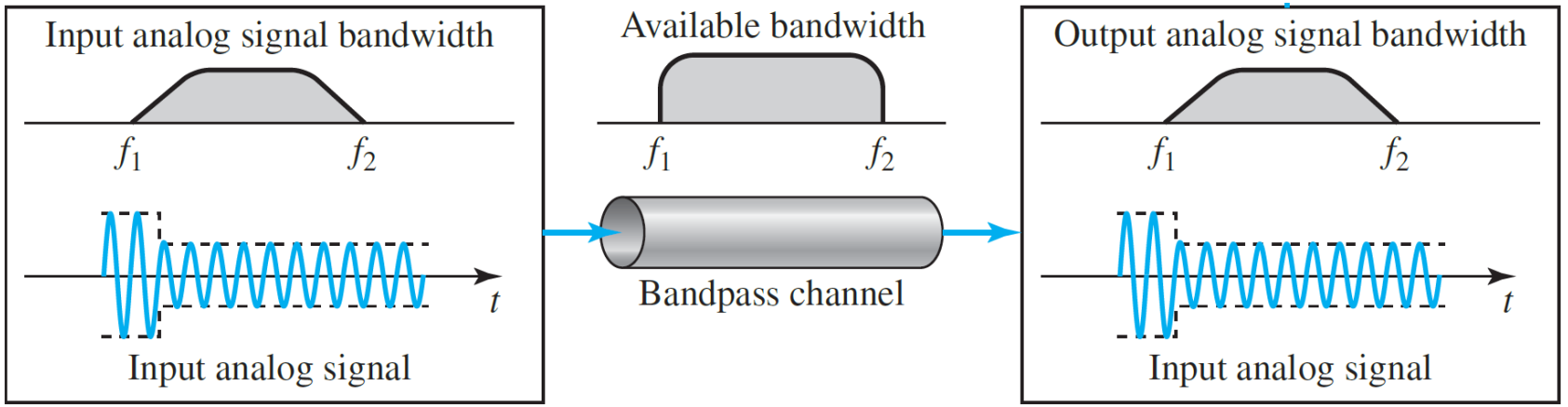
Baseband transmission

Low-pass channel supports digital transmission.



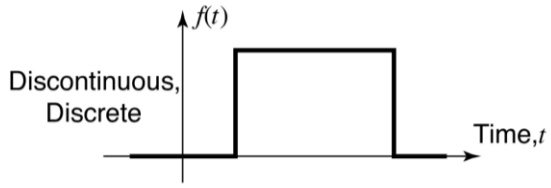
Broadband transmission

Band-pass channel supports analog transmission.

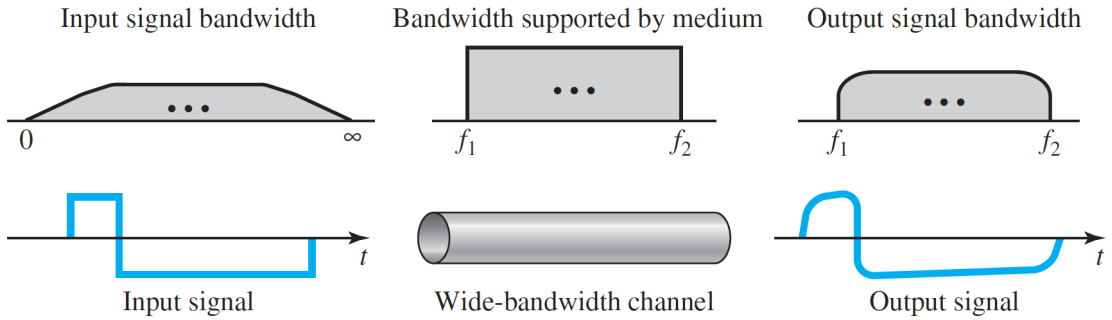


Data and Signal Conversion

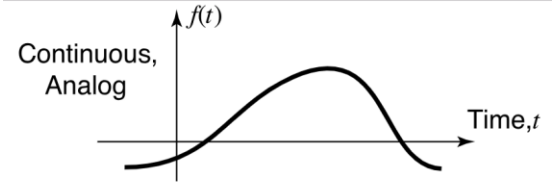
digital data input **converter**



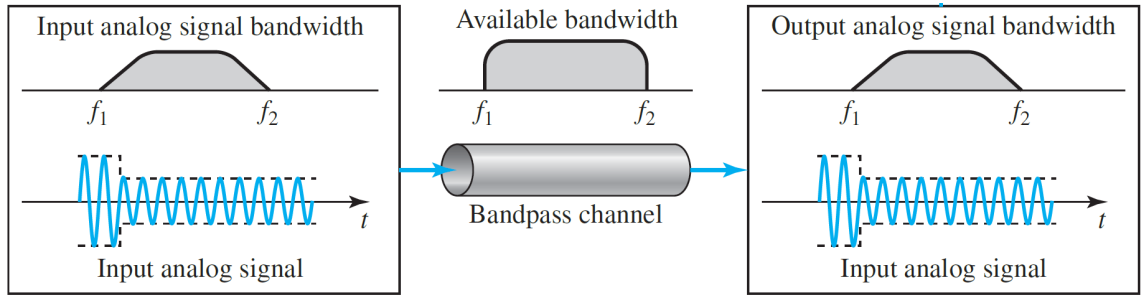
Low-pass channel supports digital transmission.



analog data input



Band-pass channel supports analog transmission.



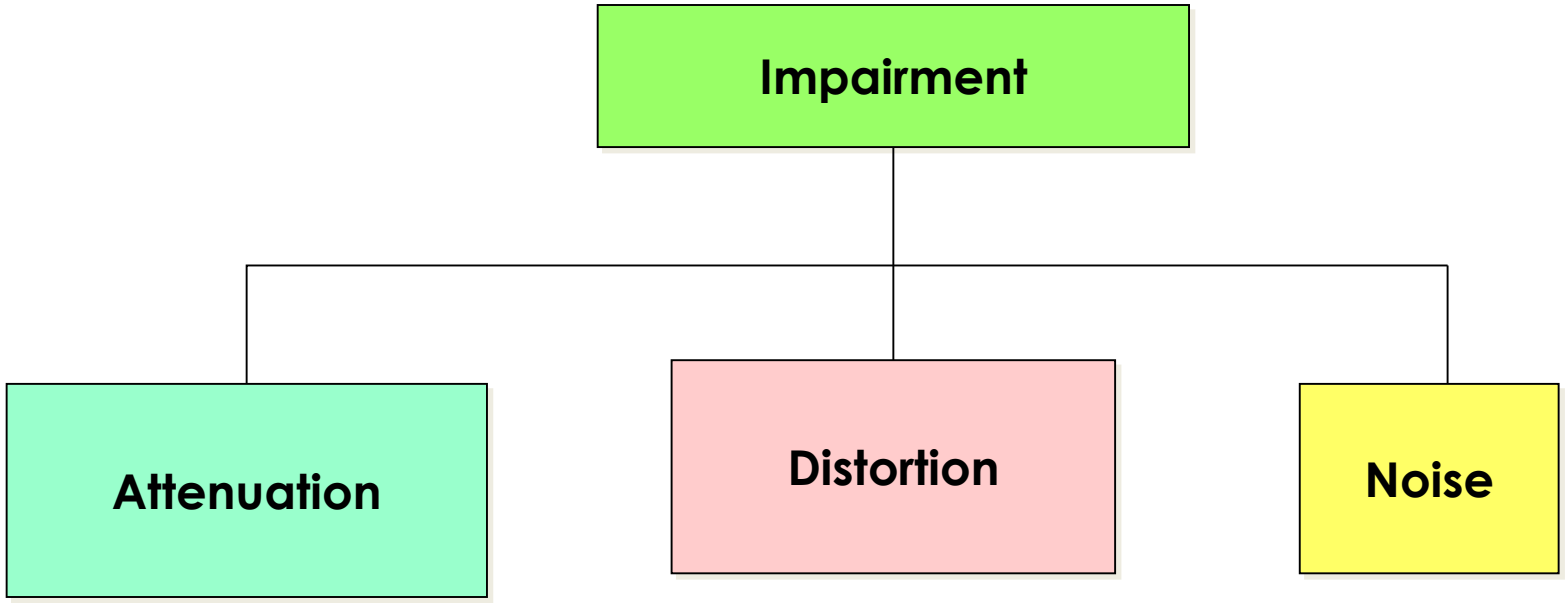
Data and Signal Conversion

	Channel	
Data	**Digital**	**Analog**
Digital	**Digital-Digital** conversion (**Line coding**)	**Digital-Analog** conversion (**Shift keying**)
Analog	**Analog-Digital** conversion (Pulse Code Modulation) (**Digitization**)	**Analog-Analog** conversion (**Modulation**)



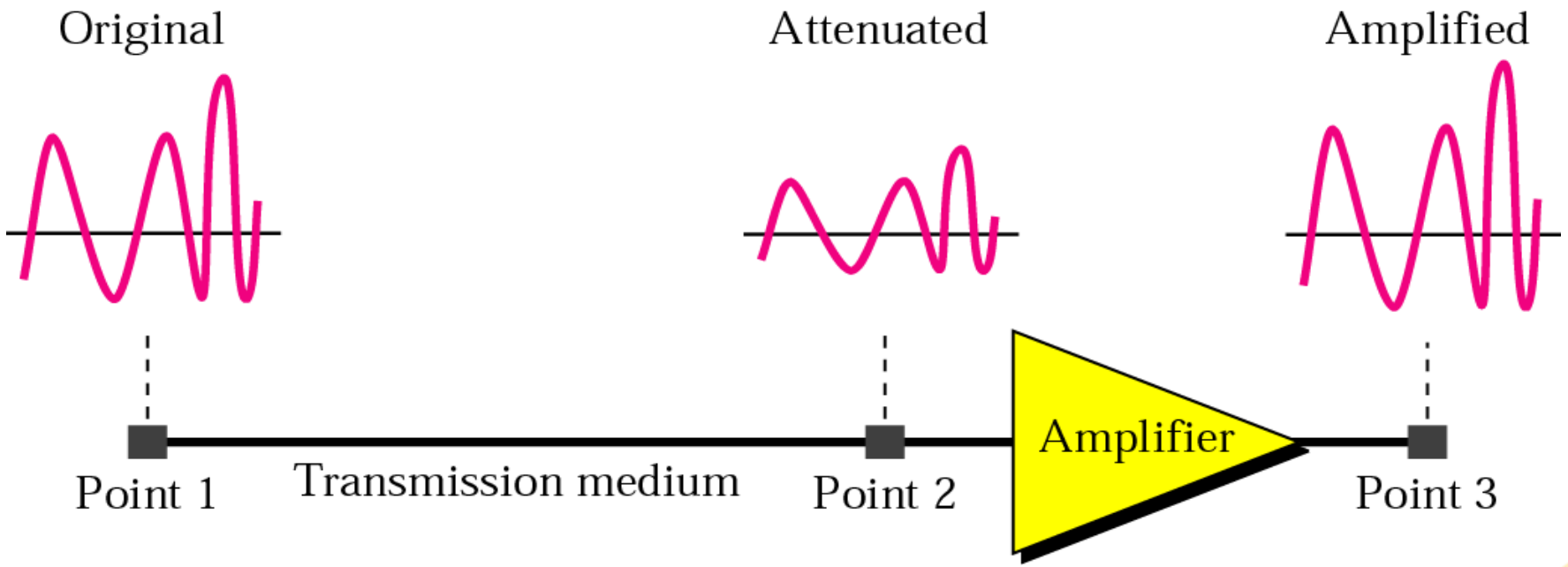
Transmission Impairment

- The imperfections in transmission media cause impairment in the signal



Attenuation

- Attenuation is the loss of signal energy due to the resistance of the transmission media.
- Attenuation limits the length of the transmission media.



Power Ratio

- The amount of attenuation can be measured as signal power ratio with respect to the original signal power. The unit of the ratio is decibel.

$$\mathrm{dB} = 10 \log_{10}\left(\frac{P_2}{P_1}\right)$$



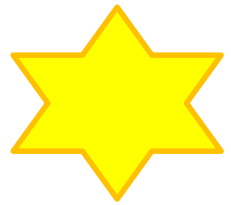
- where
  - ✓ P_2 is the signal power at point 2.
  - ✓ P_1 is the signal power at point 1.

- If dB is negative, we have attenuation ($P_2 < P_1$).
- If dB is positive, we have amplification. ($P_2 > P_1$)

Attenuation Examples

- Imagine a signal travels through a transmission medium and its power reduced to half. This means that $P_2 = \frac{1}{2}P_1$.
  - ✓ In this case, the attenuation can be calculated as

$$\text{dB} = 10\log_{10}\left(\frac{P_2}{P_1}\right) = 10\log_{10}\left(\frac{\frac{1}{2}P_1}{P_1}\right) = 10\log_{10}\left(\frac{1}{2}\right) = -3\text{ dB}$$



- Imagine a signal travels through a transmission medium and its power is increased by a hundred time. This means that $P_2 = 100\,P_1$.
  - ✓ In this case, the amplification can be caluculated as

$$\text{dB} = 10\log_{10}\left(\frac{P_2}{P_1}\right) = 10\log_{10}\left(\frac{100P_1}{P_1}\right) = 10\log_{10}(100) = 20\text{ dB}$$

Log(2) = 0.3; Log(x/y) = Log(x) - Log(y); Log (x*y) = Log(x)+Log(y)

Example

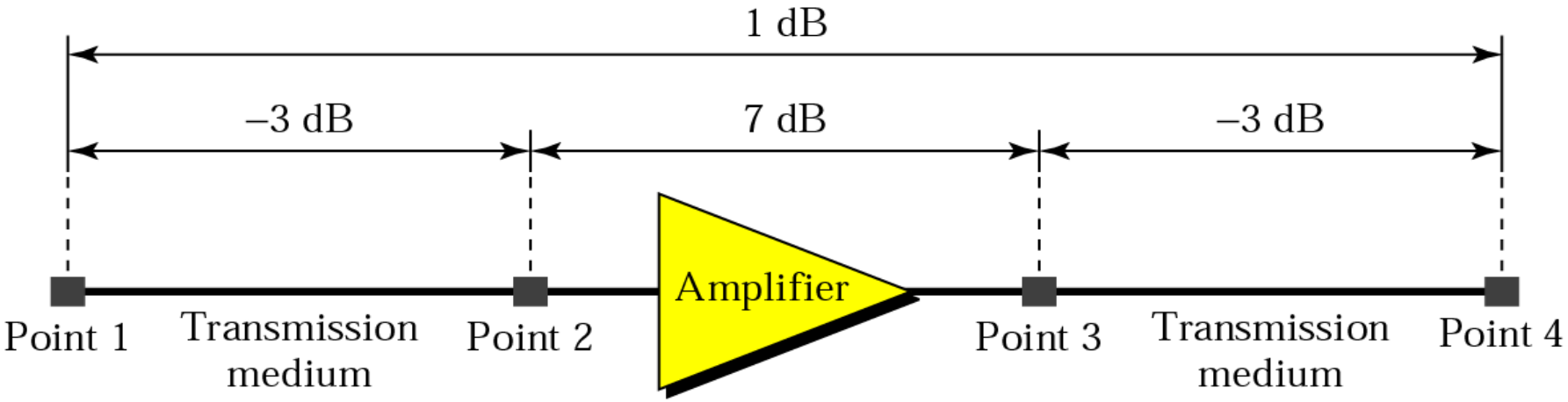
- Imagine a signal travels with 1 mW through a transmission medium and its power reduced to 0.2 mW. $How\ many\ decibels?$

$$\mathrm{dB} = 10\log_{10}\left(\frac{P_2}{P_1}\right) = 10\log_{10}\left(\frac{0.2m}{1m}\right) = 10\log(2*10\text{^}{-1})\mathrm{dB}$$

= 10log(2) + 10log(10^-1) = 3-10 = -7 dB

Attenuation Example

- What is the signal power ratio in dB between point 4 and point 1?



dBm

- Sometimes the decibel is used to measure signal power in milliwatts.
- In this case, we can represent this absolute measurement (not relative) of power in milliwatts as dBm which can be calculated as follows

$$\mathrm{dB_m} = 10 \log_{10} P_m$$

- where P_m is the power in milliwatts (Change watts to milliwatts by multiplying with 10^3).
- Example, calculate the power of a signal if its $\mathrm{dB_m}$ is -30.

$$\mathrm{dB_m} = 10 \log_{10} \longrightarrow dB_m = -30 \longrightarrow \log_{10} P_m = -3 \longrightarrow P_m = 10^{-3}\,\mathrm{mW}$$

Example

- The loss in a cable is usually defined in decibels per kilometer (dB/km). If the signal at the beginning of a cable with -0.3 dB/km has a power of 2 mW, what is the power of the signal at 5 km?

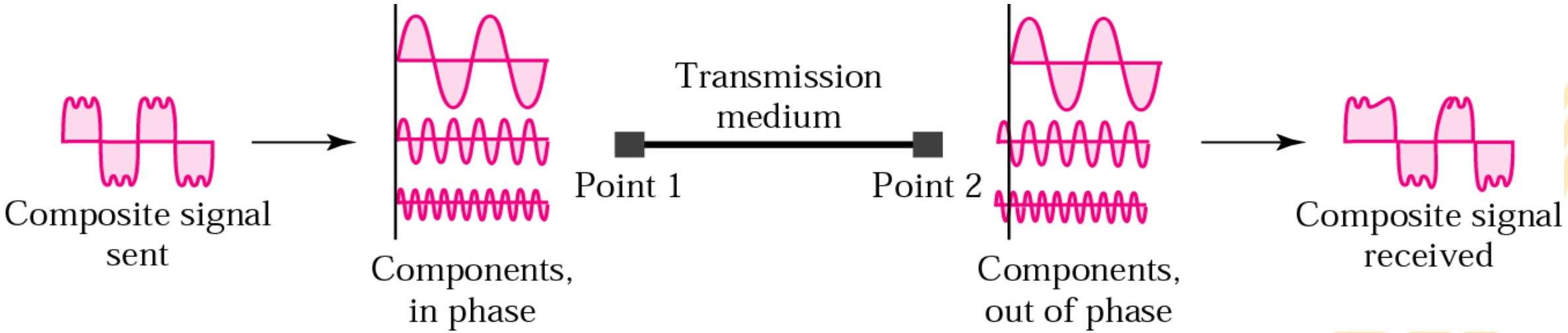
- At 5 km, the loss in the cable in decibels should be a 5 block of attenuation system. The loss should be 5 x (-0.3) = -1.5 dB. We then have

$$\mathrm{dB} = 10 \log_{10}(P_2/P_1) = -1.5 \longrightarrow (P_2/P_1) = 10^{-0.15} = 0.71$$

$$P_2 = 0.71P_1 = 0.7 \times 2\,\mathrm{mW} = 1.4\,\mathrm{mW}$$

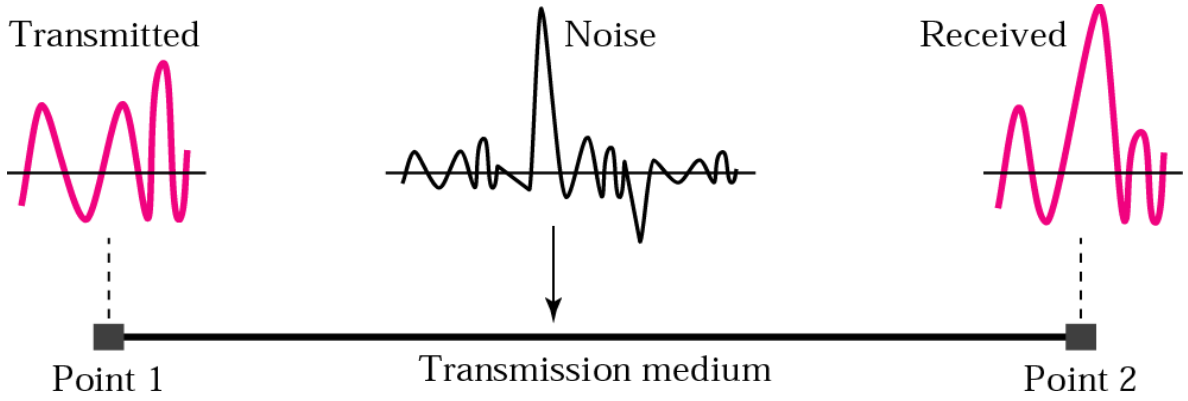
Distortion

- Distortion occurs in a composite signal, especially digital signals, which is made of different signals with different frequencies.
- When the signal components of a composite signal arrive at the destination, the characteristics of the recombined signal may change its shape due to various delays of signals.



Noise

- Noise are external signals coming from the environment which can interfere with the transmission of signals carrying data.

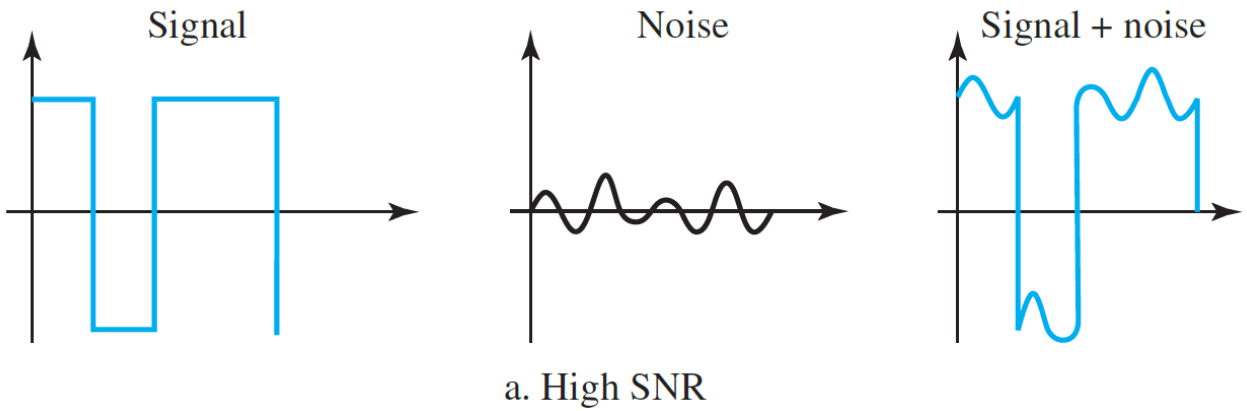


- There are 4 types of noise
  - ✓ Thermal noise (from randomly moving electrons)
  - ✓ Induced noise (from external magnetic field change)
  - ✓ Crosstalk (from magnetic/electrical field change from nearby wires)
  - ✓ Impulse noise (from a spike of energy change, e.g., thunder)
- We use **Signal-to-Noise ratio (SNR)** to measure the effect of noise to the signal

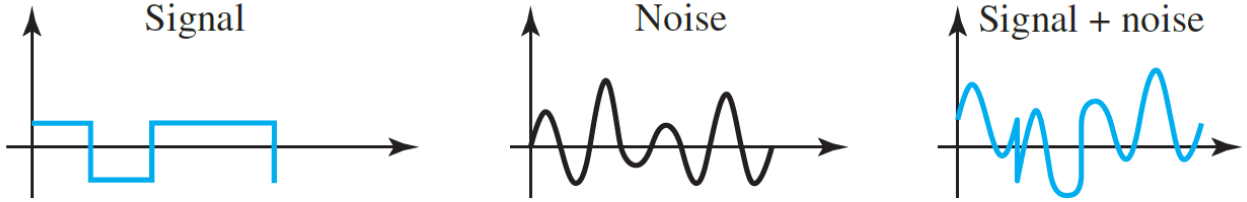
Signal-to-Noise Ratio (SNR)

- Signal-to-Noise Ratio (SNR) is a measure of the impact of noise to the signal. The signal to noise ratio (SNR) is defined as

$$\text{SNR} = \frac{\text{average signal power}}{\text{average noise power}}$$



a. High SNR



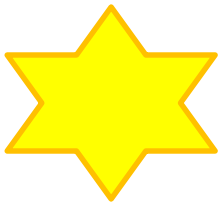
b. Low SNR

Signal-to-Noise Ratio (SNR)

- SNR is actually the ratio of what is wanted (signal) to what is not wanted (noise).
  - ✓ A high SNR means the signal is less corrupted by noise;
  - ✓ a low SNR means the signal is more corrupted by noise.

- Since SNR is the ratio of two powers, it can also be described in decibel units.

$$SNR_{dB} = 10 \log_{10} SNR$$



Signal-to-Noise Ratio (SNR)

- **Example 1**: the power of a signal is 10 mW and the power of the noise is 1 μW; what are the values of SNR and SNR_{dB}?

- Solution

$$\text{SNR} = (10{,}000\ \mu\text{w}) / (1\ \mu\text{w}) = 10{,}000 \qquad \text{SNR}_{\text{dB}} = 10 \log_{10} 10{,}000 = 10 \log_{10} 10^4 = 40$$

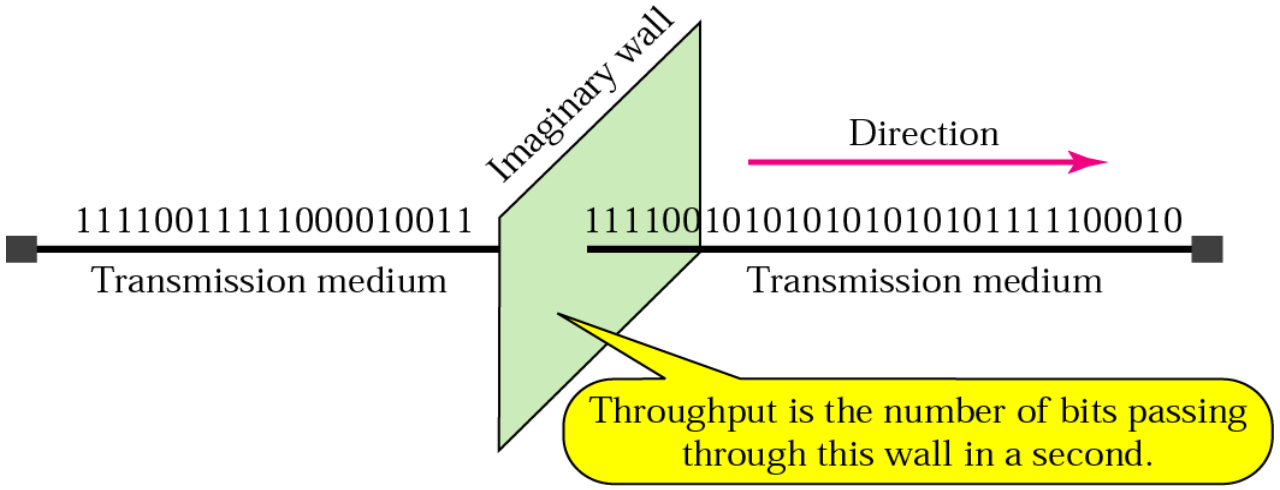
- **Example 2**: the values of SNR and SNR_{dB} for a noiseless channel are

- Solution

$$\text{SNR} = (\text{signal power}) / 0 = \infty \longrightarrow \text{SNR}_{\text{dB}} = 10 \log_{10} \infty = \infty$$

Signal Characteristics

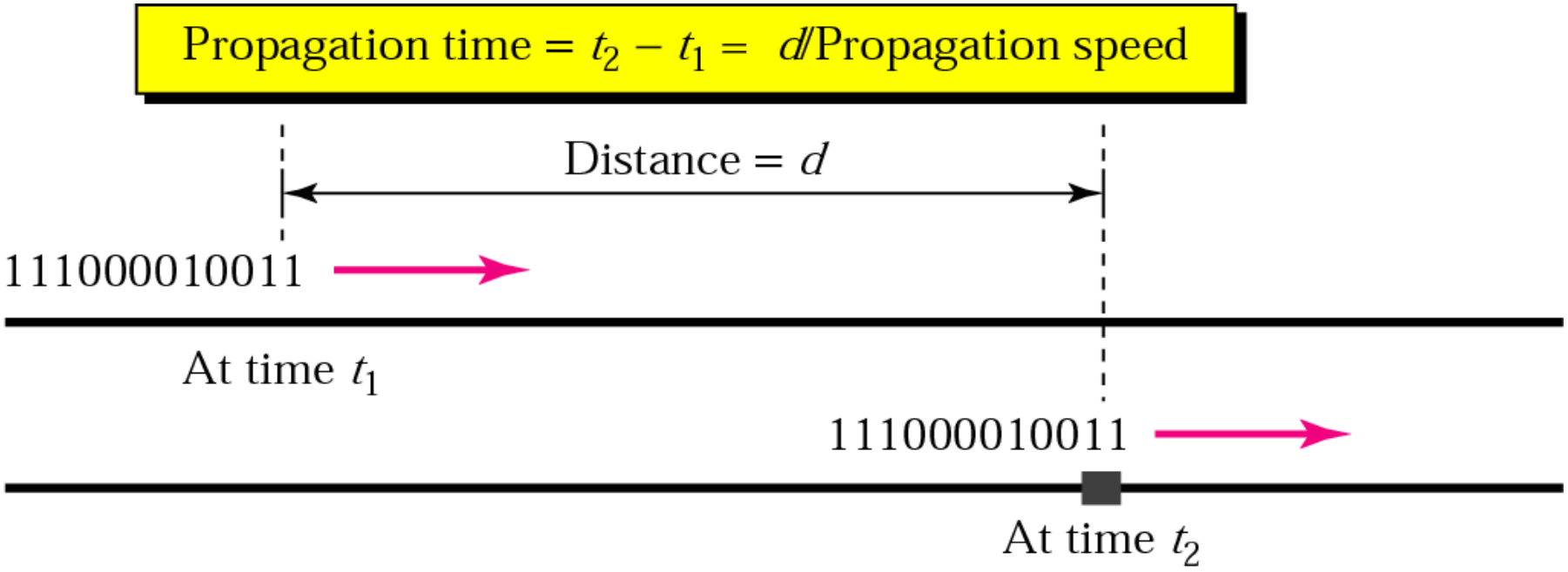
- Throughput: the measurement of how fast data can pass through an entity (a point or a network). Unit is bits per second.



- Propagation speed: the distance a signal or a bit can travel through a medium in one second (meter per second)

Signal Characteristics

- Propagation time: the time a signal or a bit can travel through a medium in a designated distance.



Signal Characteristics Examples

- A network with bandwidth (maximum bitrate capacity) of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the **throughput** of this network?

$$\text{Throughput} = \frac{12{,}000 \times 10{,}000}{60} = 2 \text{ Mbps}$$

- The throughput is almost 1/5 of the bandwidth in this case.

Signal Characteristics Examples

- What is the **propagation time** if the distance between the two points is 12,000 km? Assume the propagation speed to be 2.4×10^8 m/s in cable.

$$\text{Propagation time} = \frac{12{,}000 \times 1000}{2.4 \times 10^8} = 50 \text{ ms}$$

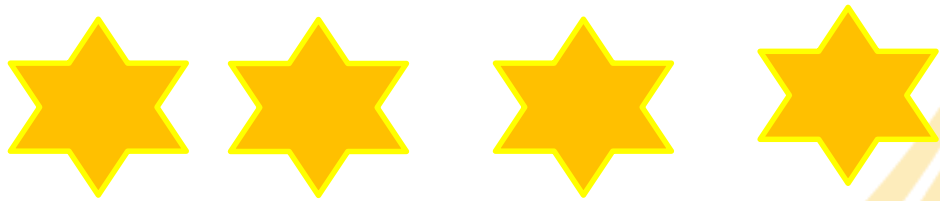
- The example shows that a bit can go over the Atlantic Ocean in only 50 ms if there is a direct cable between the source and the destination.

Data Rate Limit

- How fast we can send data?

- Data rate depends on 3 factors
  - ✓ Bandwidth available of the channel
  - ✓ Level of signals we use
  - ✓ Quality of medium/channel (level of noise in the channel)

- In order to calculate data rate appropriate for each channel, we consider
  - **Nyquist bit rate** for noiseless channel
  - **Shannon bit rate** for noisy channel

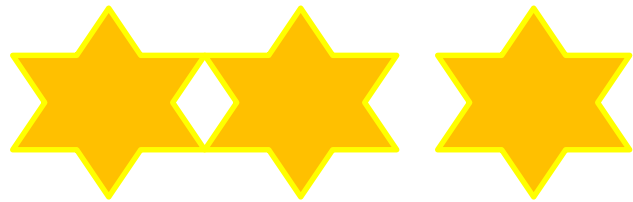


Nyquist Bit Rate

- Nyquist bit rate is the theoretical maximum bit rate for the channel.
- Assume there is **no noise** (=noiseless) in the channel.

$$\text{BitRate} = 2 \times \text{bandwidth} \times \log_2 L$$

- Bandwidth is the bandwidth of the channel
- **L** is the number of **signal levels** used to represent data
- BitRate is the bit rate in bits per second



- Note: the more levels we use, the more difficult it is for the receiver to distinguish the signal voltage of each level.

Nyquist Examples

Example 3.34

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$$

Example 3.35

Consider the same noiseless channel transmitting a signal with four signal levels (for each level, we send 2 bits). The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 4 = 12{,}000 \text{ bps}$$

Extra: If the bandwidth is 2 KHz and we transmit 8 signal levels, what is the maximum bit rate?

Nyquist Examples

Example 3.36

We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need?

Solution

We can use the Nyquist formula as shown:

$$265{,}000 = 2 \times 20{,}000 \times \log_2 L \longrightarrow \log_2 L = 6.625 \longrightarrow L = 2^{6.625} = 98.7 \text{ levels}$$

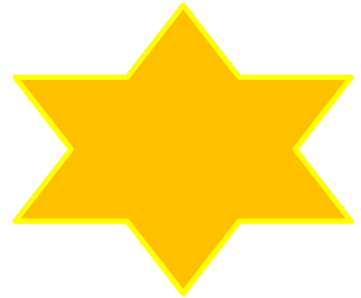
Since this result is not a power of 2, we need to either increase the number of levels or reduce the bit rate. If we have 128 levels, the bit rate is 280 kbps. If we have 64 levels, the bit rate is 240 kbps.

Extra: If the bandwidth is 1 KHz and we want the max bit rate of 10 kbps, how many signal levels ?

Shannon Bit Rate

- In reality, the channel is always **noisy**. We must take into account, the level of noise in the channel.

$$\text{Capacity} = \text{bandwidth} \times \log_2(1 + \text{SNR})$$



- Bandwidth is the bandwidth of the channel.
- SNR is the signal level to noise level ratio
- Capacity is the capacity of the channel in bits per second.

Shannon Examples (1)

Example 3.37

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity C is calculated as

$$C = B \log_2 (1 + \text{SNR}) = B \log_2(1 + 0) = B \log_2 1 = B \times 0 = 0$$

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot receive any data through this channel.

Example 3.38

We can calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal-to-noise ratio is usually 3162. For this channel the capacity is calculated as

$$C = B \log_2 (1 + \text{SNR}) = 3000 \log_2(1 + 3162) = 3000 \times 11.62 = 34{,}860 \text{ bps}$$

This means that the highest bit rate for a telephone line is 34.860 kbps. If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to-noise ratio.

Shannon Examples (2)

Example 3.39

The signal-to-noise ratio is often given in decibels. Assume that $SNR_{dB} = 36$ and the channel bandwidth is 2 MHz. The theoretical channel capacity can be calculated as

$$SNR_{dB} = 10 \log_{10} SNR \longrightarrow SNR = 10^{SNR_{dB}/10} \longrightarrow SNR = 10^{3.6} = 3981$$

$$C = B \log_2(1 + SNR) = 2 \times 10^6 \times \log_2 3982 = 24 \text{ Mbps}$$

Terminologies after the break

1. Low Pass Filter
2. Band-Pass Filter
3. Digital Transmission
4. Analog Transmission
5. Attenuation
6. dB equation
7. dBm equation



8. Distortion
9. Noise
10. SNR equation
11. Propagation Time
12. Throughput
13. Bandwidth-Delay Product
14. Nyquist Bit Rate **
15. Shannon Bit Rate **

ITCS217 DataCom
Quiz #1 (Lectures 1 & 2 & 3)

Thu 27 & Sat 29 August 2026

Be honest and Do it by Yourself!

Sec1 & 3 : Thu 27 August 2026, 13:00-13:15

Sec2: Sat 29 August 2026, 14:00-14:15

Paper-Based
Prepare a pen

